# 1703 年 5 月 18 日 莱布尼茨致白晋的一封信

（上海海洋大学 李婷玉 译 ）

我极尊敬的神父，

您尊贵的信件，日期为北京 1701 年 11 月 4 日，由极尊敬的神父郭弼恩转交，于 1703 年 4 月 1 日送达我处。我很高兴得知您尊贵身体健康，以及欧洲人在中国的事务进展顺利，因为这关系到基督教宗教的进步。[1] 但我不能免除自己敦促您和您的同伴，要比迄今为止更充分地利用有利时机，以便我们以我们的知识交换中国人的精妙知识，尤其是在技艺方面，因为您信中的一段话让我相信，当他们认为自己不再那么需要外国人时，他们对外国人的善意有一天会变得极其冷淡。这段话就是您所讲述的，我尊敬的神父，即皇帝几年来已经达到他所希望了解的我们科学理论的满意程度，不再给予欧洲人同样的机会与他就这些事务交谈。既然如此，在我们还有手段的时候，刻不容缓地研究中国技艺的历史极为重要，而且尤其如此，因为他们学习我们的科学和知识比我们学习他们的更容易，既因为我们的更多在于推理而他们的更多在于经验，也因为我们的大多数是公开的，而中国人的则几乎仅为专业人士所知，并且在那里是以传统方式[即私人、口头传承]流传的。[2] 我相信，您可以巧妙地让皇帝本人对这项技艺研究产生兴趣，让他意识到完善他国家技艺的方法就是为它们提供描述；通过这种方法，人们将能够看到中国人缺少什么，以便借助欧洲人的帮助来弥补不足。同样，拥有一部完整的字典，或者更确切地说几部合起来构成一部完整的字典，其中包含所有特别属于各行业的字符，而且对鞑靼人而言，重要的是中国人不要对他们隐瞒任何东西。现在，关于术语，当涉及解释它们时，应提供自然物和人造物的插图以及方法或秘密的描述。特别地，我尊敬的神父，我请求您为我尽可能清晰地获取关于所有种类中国纸张的制造以及装饰它的方法的描述。他们用来印刷字符的字模[即印刷字体]是用什么制成的？从前，年轻的范·赫尔蒙特先生[3] 告诉我，他们是用一种漆或清漆制成的，在它还软的时候，他们很容易在其中刻出这些字符，但之后它会像木头一样硬化；那会很方便。我还希望了解他们是否有好的方法来强化皮革或织物，使空气和水

完全无法渗透进去，尤其是空气，因为我尚未在欧洲满意地发现有人在这方面取得成功，以便制造可以充气并在上面睡觉的大型袋子或垫子。

至于欧洲的事务，它们处于这样一种状态，以至于让我们羡慕中国人。西班牙国王的去世引发了一场可怕的战争，这场战争将决定欧洲的命运。[4] 沙皇与瑞典国王之间的战争仍在继续。[5] 但后者[即瑞典国王]，非但没有击退莫斯科人，反而将他的军队转向波兰，并进入这个王国的腹地，莫斯科人重新获得活力，并大大骚扰与他们相对的瑞典省份，甚至夺取了诺特堡，一个具有一定重要性的地方。[6] 他们还试图沿黑海一侧扩展，建造一座新要塞，如果苏丹在经历一场奥斯曼帝国遭受如此巨大损失的战争后不想不惜一切代价维持和平，[7] 并且如果他想同意克里米亚鞑靼人的请求，[8] 他或许已经再次与莫斯科人开战，对莫斯科人而言，其目标似乎是获取更多波罗的海港口；但他们会发现这很困难。瑞典国王迄今似乎远未与波兰国王达成和平，他本希望将后者赶出他的王国，因为他实际上在波兰人和立陶宛人中有一大派系支持这种倾向。但事务似乎已转向另一个方向，而且看起来瑞典国王最终将能够决定与国王和共和国打交道，在那里他希望看到社会反对莫斯科人。然而，我不相信波兰国王会对此提供帮助，因为沙皇忠实地帮助了他，而且几年前波兰人与莫斯科人之间缔结了永久和平，据我所知，波兰人完全没有理由打破它。

但让我们放下那些几乎不涉及中国的世界事务，回到科学上来。无穷小分析法为我自己以及那些推进我方法的人提供了手段，来解决几何学此前未能解决的困难。例如，没有人确定过伽利略相信是抛物线的悬链线的图形。现在，我已经证明，通过一根非常细的链条，像时钟中使用的那些，悬挂在其两端，人们将无需计算即可获得现成的对数，其精确度取决于链条的大小、细度和柔韧性。[9] 我也不知道我是否曾向您谈过我的动力学，或力的科学，在那里我找到了确定它们的完美方法，从实在形而上学的崇高原则先验地推导出来，[10] 而且进一步完全符合实验，在这些实验中我不仅发现笛卡尔派相信的运动量守恒是错误的，而且绝对力（按其应被理解的方式）是[所]守恒的；而马里奥特先生、德拉伊尔先生以及最近的其他人仅[提出]一种相对于另一物的[运动]守恒。我还发现，在任何一个时刻都有与任何其他时刻同样多的作用，无论是在宇宙中还是在被设想为仅在它们自身之间有接触的物体中。但运动作用量与笛卡尔派所称的运动量完全不同。[11] 此外，我证明力

是物体实体的本质，而且它是古人的隐德来希，尽管它需要由物体的巧合或由上帝的最初配置来确定。因此，正是通过对力的考虑，我为古人辩护，反对现代人过于物质的哲学，我承认在物体中一切都机械地运作，但我向他们表明，他们没有充分考虑到力学本身的原则来自某种高于仅仅物质或受想象支配的东西。

但让我们来谈您信件的主要主题之一，即我的二进制算术与伏羲模型的关系，伏羲被认为是世界上最古老的国王和哲学家之一，并被视为中华帝国的创始人和他们科学的创始人。因此，这个模型是今天在宇宙中发现的最古老的科学纪念碑之一，看起来已有 4000 多年历史，而且或许几千年来完全未被理解，这是一件非常令人惊讶的事，它与我的新的算术方式完美匹配，而且我恰好在适当的时候写信给您谈我的算术，也就是说，正当您致力于破译这些线条的时候。[12] 我向您保证，如果我没有制定我的二进制，我自己或许会长时间注视这个伏羲的卦或线条系统而无法洞察其意义。因为我脑海中已有二十多年这个由 0 和 1 组成的算术，在其中我预见到将数字科学推进到超越我们迄今为止为它所达到的任何完美程度的奇妙手段；但我一直克制不提出它，直到我准备同时展示其巨大优势。然而，由于一千种其他职业和沉思阻碍了我研究它，我最终披露了它（尽管尚未在任何印刷书籍中），以便这样一个具有如此后果的思想不会丢失。而且我很高兴[我这样做]恰恰是在您能够如此善加利用它来破译这个中国古代纪念碑的时候。实际上，似乎有某种天意的指引。而且我毫不怀疑，当这个算术进一步发展时——如果有一些既有能力又有意愿致力于此的人，它很快就会发展——它将成为重新唤醒皇帝和中国领导人物注意力的新手段，尽管您对伏羲卦的破译本身就足以令他们震惊。

我发现正方形与伏羲的圆形是同一事物，甚至有助于解释它，因为圆形中的秩序有点混乱。我在圆形中从右下侧开始，也就是从 和等等，或从 0 和 1，或更确切地说从 000000 和 000001 开始。因为必须将模型视为眼睛位于中心，并从靠近中心的线条开始。这也由字符显示出来，因为它们从中心看时就是这样，或者更确切地说，如果将圆形转动，使字符中离中心最远的部分成为最上方；这就证明了与正方形字符的一致性，这些字符因此显得相同，以至于就圆形而言，既没有高也没有低，正如就地球的球体而言，离中心最远的被视为最高；而且甚至可能伏羲考虑到了这一点。但您无论如何是正确的，我极

尊敬的神父，相对于这个圆形内所刻的正方形，写 ἀνω [ano，即上方、上、顶] 和 κάτω [kato，即下方、下、底]。[13]

但回到圆形的卦或线性图形，从右下侧的 0 和 1 开始，我们向上行进到右上侧，也就是向上到（最靠近中心的线条始终被视为第一个），即 011111 或 31。然后（而不是继续在圆形中并经过左上侧然后下降）我们从左下侧重新开始，以便从那里向上；现在，在左下侧的是 ，即 100000 或 32，显然是因为这样 0 和 32+0，或 1 和 32+1，或 2 和 32+2（用二进制表示仅在于一个线性图形以□或 0 开始，而另一个以□或 1 开始）在从下到上上升时面对面。或许，也希望这能提供一点谜题，并显示圆形与正方形之间的一些表面差异。最后，从左下侧向上到顶，我们以 ，即 111111，或 63 结束。至于所刻正方形，那里的秩序是完全自然的，从

000000 000001 000010 000011 等,
0 1 2 3

开始，其中应注意到数字的秩序在每行或每排数字中从左到右进行，而行或排从上到下跟随；在每种情况下，这都符合我们在欧洲写作的惯例，与东方[民族]和中国人的习俗相反。然而，确实，在每个数字中必须从下到上计数线条，或许是为了模仿圆形的对应关系，在那里我们从下到上开始，从最靠近中心的开始。这个如此古老的纪念碑与欧洲写作的对应让我想起我读过，在一些非常古老的东方纪念碑中，似乎也以欧洲风格阅读。这如果我没有弄错的话，就是在波斯波利斯或切希尔米纳尔废墟中发现的奇特文字，[14] 那里文字由线条和三角形组成，而且很可能早于波斯君主制。[15] 这个古代纪念碑在西班牙驻波斯大使菲格罗亚的记述中被提及，[16] 以及在已故 M. 泰弗诺编辑的旅行关系[集]中，[17] 以及其他地方。

但让我们回到仅使用无和一的二进制演算。您敏锐地感觉到它对宗教的一个主要用途，即它是创造的强有力象征，[18] 即万物仅从上帝和无中起源，完全没有任何预先存在的物质；[19] 而且非绝对而是相对的无——即它的限制——本质上存在于受造物中，与它们的

缺陷成比例，这种限制无非是对纯现实或纯作用的进一步发展的否定，正如一个圆被其圆周所束缚，从而赋予它其最终定义。而且我相信，当中国的博学者很好地掌握这个观念，尤其是当他们看到伏羲的巧妙设计符合我们自己的[传统]时，他们将充分倾向于相信这位伟人还希望代表上帝、万物创造者，以及他从无中引出那些事物的创造。因此，这可能是您的教理问答的最重要要点之一，取自中国经典作者，并值得向皇帝本人解释。[20]

至于数字的二进制表达，即用 1 和 0，用于数字科学的完善，我有论证可以说服我们，这是一种将这门科学远远超出其先前状态的方式；因为我发现自然数的平方、立方和其他幂也在其列中具有[重复]周期，正如自然数一样，这为确定它们并轻松建立它们的表格以及发现迄今未知的关系提供了奇妙的便利。[21] 我还发现，这在用趋向无穷的整数序列表达不可公度的几何维度方面将具有巨大效用，而在鲁道夫·范·屈伦[22] 的圆周计算中（例如）所缺少的是，没有一个公式可以继续下去。但由于我完全没有闲暇进一步研究这些，如果我找不到有能力或意愿协助我的人，我将把从中可以收获的好处留给后代，这些好处比仅考虑事物表面的人所能想象的范围更大。

关于您尊贵也提到的用于观念排列的双重几何级数的用法，经验让我意识到二分法在形成观念方面的巨大用途。[23] 确实，人们可以用不同方式进行二分法，但它们通向同一目的，即通过不同的下属类别达到同一物种。而且正是这一点使得同一术语可以接受不同的定义，然而分析最终将表明它们是一致的。我相信，很少有人比我更多地研究过观念的排列，并且为它们制定了更大的方案，但在职业和分心之中，我无法想象如何在没有比我更有活力和闲暇的人的协助下清理我的混乱。然而，如果上帝赐予我多几年的生命，我将努力为这项工作带来一些秩序，在我看来这是人们可以承担的最重要的工作之一，因为它将为推理提供一条阿里阿德涅之线，即一种以演算形式的有形方法来引导我们，无论是在判断还是发明的事务中。[24]

我怀疑伏羲将 64 个数字（要么未复合或扩展到 128 甚至更远）分配给他构想的最基本的术语，并且为每个这样的术语提供了它自己的字符，该字符也标示其数字或等级；然后从这些最简单和主要的术语和字符，他通过添加小笔画形成了其他字符。但随着时间的

推移，这些字符被改变了，既由于大众用法的性质逐渐改变笔画（正如在比较某些语言的古代文字与现代文字时可以看到的），也由于那些不再知道字符的逻辑或方法的人，根据他们自己的奇想调整它们，这些奇想往往基于隐喻或其他更微弱的联系。如果我们很好地理解中国文学史——我在写信给您时曾希望看到它按照好的批判方法建立起来，以便区分古代与现代——那么我们将能够更好地做出判断。也很可能不同的君主或哲学家对字符进行了不同的改革，以使它们更好；但由于他们并不总是遵循相同的语法或词源法则，可以说，他们的起源最终完全被掩盖，甚至比伏羲的断线的情况更甚，在那里意义已经丢失但刚刚被我们恢复。[25] 我将非常高兴地了解附在伏羲模型中每个数字上的中国字符的意义，如果不会给您带来太多麻烦，我恳请您尊贵将解释寄给我。显然它们是现代的解释或评论，是在不知道模型的真正用法的情况下编造的，即用 0 和 1 的演算。[26]

如果您发现今天的中国人完全不知道这个演算，那么您尊贵或许可以在皇帝本人和中国主要学者在场的情况下，大胆地将成功破译伏羲模型及其断线的真正意义归功于来自欧洲的新发现，即用 0 和 1 计算的方式。在我看来，这不会被视为无关紧要，而且应该提高中国人对欧洲科学的尊重，并因此尊重我们的宗教。这也将使他们对其中仍隐藏的尚未被发现的神秘充满巨大期待，而且甚至给我们提供一个开放的领域，来设计一种新的特征，它将被视为伏羲[模型]的扩展，并将成为观念分析和这个我有一个初步计划的奇妙理性演算的起点。这个秘密而神圣的特征还将给我们提供手段，向中国人介绍哲学和自然神学的最重要真理，以便为通往启示[神学]铺平道路。而且无论它与他们自己的多么新或不同，它将因伏羲而被接受，并最终成为一种仿佛仅属于最高等级学者、最开明且与伏羲最紧密相连的语言，甚至值得单独的[研究]学会和学院。因此，如果人们知道如何利用它，无论是在您现在所在的地方还是更多地在欧洲，这个发现可能对中国整个帝国产生巨大影响。[划掉的部分，查原稿第 33a 页]

因为现在我们已经找到了这些线条的钥匙，而且中国人完全不能怀疑我们已经成功了，我们将在他们眼中获得无比更高的信誉。我们可以以此方式建立的特征将把观念还原为数字，同时具有将它们从属于演算的优势，正如数字一样，这是我们迄今为止在这些事务中所承诺的一切之外的，而且许多人并非没有一些似是而非的理由，将相信这是不可能

的。而且还需要一些呈现手段，因为这是某种不容易想象的东西，而且它应该携带它自己的证明。您将判断，我极尊敬的神父，这是否不应该激励那些关心人类精神完善的人，尤其是关心美德和真正宗教进步的人。但目前的欧洲战争，以及或许这个一般特征发现的重要性和巨大规模，大大减少了我看到它在必要协助下实现的希望，因为人们想知道天意是否不希望为另一个更适合吸引上天恩典的时代保留它。

您将由此看到，我极尊敬的神父，仅仅设置几个划分和子划分以达到[事物的]物种是不够的，因为这完全无法提供一种能够通过这些划分赋予它们的特征来提供那些物种属性的分析，在其中还必须在“混乱的”观念和清晰的观念之间做出重大区分。[27] 因为清晰概念的特征可以从一开始就精确，但混乱[概念]的特征只能是暂时的，并且与已被承认为清晰的那部分成比例。留给后代根据新实验来完善它们。[28] 而且这个观察很重要，既是为了不承诺超出人类当前力量的东西而让我们暴露于虚荣，也是为了准确知道有什么要做的。至于颜色，棱镜实验表明，除了白和黑，主要颜色是黄、蓝、[和]红——红与蓝混合成紫——而且绿只是黄和蓝的组合；而且[这些实验还表明]这是由折射期间光线的弯曲决定的，而不是仅由黑和白的混合决定的。[29] 但这只是顺便说说，而且您仅引用颜色作为澄清的例子，在那里适用的规则是 exemplorum non requiritur veritas。[30] 我也不想从第三度跳到第六度，因为方法既不是很好确立的，也不会容忍这类跳跃而不受损害。经院学者赋予性质的八个度（因为我不记得亚里士多德使用过它们）[31] 来自他们相信四个元素中的两个始终具有共同性质，并希望在每个中放置四个度；例如火中的四个热度（按照化学家的说法）以及空气中的另外四个。

一个名叫苏塞特[32] 的英国人，被称为卓越的计算者，生活在 300 多年前，以极端的程度完善了强度和缓和的概念，即性质内的度。但这种计算方式似乎完全不适合我们的字符。事物以及观念不仅 gradu tantum [即仅在程度上]不同，而且在许多其他方面不同。

然而，似乎中国人认为基本的八个 卦，或八个线性图形，可以让人相信伏羲本人在制作一切来自一和无时心中有创造，而且他甚至将这种对应扩展到创世记的故事。因为 0 可以表示创造天地之前的虚空，然后跟随七天，每一天标示那天开始时已存在并被发现已

造的东西。在第一天开始时存在 1，即上帝。在第二天开始时，2，天地在第一天被创造。[33] 最后，在第七天开始时已经存在一切；这就是为什么最后一天是最完美的安息日，因为在那一天一切都被发现已造且充满；因此 7 是用 111 而不带 0 书写的。而且只有在这种用 0 和 1 书写的方式中，才能看到被视为神圣的七进制数的完美，在那里甚至更值得注意的是，它的字符与三位一体有一些对应。[34]

我相信我以前曾写信告诉您，一位来自柏林的神学家，几年前去世，名叫安德烈亚斯·米勒，波美拉尼亚人，在东方文献和一般语言知识方面相当博学，并且为马可·波罗的《契丹》提供了一些极好的注释，[35] 他相信他已经找到了中国字符的某种钥匙，根据他在一份小印刷纸上发表的内容；但众所周知，他要求一大笔钱来披露它。我把他介绍给极尊敬的神父闵明我[36]，他希望在从维也纳去波兰时与他交谈，但米勒先生没有去赴约。他是一个很有功绩的人，但他同样任性。已故的勃兰登堡帝国选帝侯[37] 似乎愿意给他任何他想要的东西，但他完全不可通融，甚至从柏林撤回到他的家乡。他威胁在死前烧毁他的文件，据说他遵守了诺言。他似乎完全不知道隐藏在伏羲 卦 中的演算；他也不是一个足够的数学家来察觉这样的算术。然而，我无法轻易说服自己，这样一个有功绩的人会夸耀一个秘密，除非他认为它至少有一些可能性。因此我怀疑他瞥见了中国普通数字字符与它们的事物根字符之间的某种关系，或许还有[一些]关于小添加[即笔画]的观察，这些笔画修改根字符以形成派生字符，或或许甚至复合字符。但这是您尊贵和尊敬的神父卫方济将比我更好地判断的东西。因为我需要有好的中国字符字典，并附有欧洲解释，以便思考它。

您尊贵告诉我，在您所在的地方找不到这样的字典。但一定是有人向您隐瞒了它们，因为它们肯定存在，甚至如我回忆的那样是印刷的。我的证据基于无与伦比的戈留斯[38] 的东方手稿目录，大约六年前在荷兰拍卖出售，令我大为惊讶和遗憾。因为它们是值得最伟大国王的公共财富，而且它们本应由莱顿大学在国家元首或东印度公司的帮助下保存。现在，在这个目录中，我记得有两本中国字典，一本是中文-西班牙文，[40] 我相信是在菲律宾印刷的，而且我不记得另一本是葡萄牙文还是佛兰芒文。我希望我保留了这个目录，当我回到汉诺威时，我将努力找到它，以便清楚地向您指示一切。其中之一肯定解释了字

符，但我不能说它们是否都解释了，或者其中之一或许仅涉及中国语言。我被告知其中大部分件已去了英国。

我不知道该如何说埃及的象形文字，而且我很难相信它们与中国人的有任何关系。因为在我看来，埃及字符更常见，并且太像可感知的事物，如动物和其他东西，并因此像寓言，而中国字符或许更哲学化，似乎基于更智性的考虑，如可能来自数字、秩序和关系；因此，它们只是不相连的笔画，并不指向与任何特定种类事物的任何相似性。我知道有些人相信中国人是埃及人的殖民地，基于字符的假定一致性，但这完全没有可能性。[41] 有一本关于埃及象形文字的古书，由一位霍拉波洛所著。[42] 人们也可以在阿米安·马尔切利努斯[43] 中看到，如果我没有弄错的话，仍在罗马发现的一根方尖碑字符的解释。皮埃里乌斯·瓦莱里亚努斯，[44] 基歇尔神父在他的《埃及的俄狄浦斯》、《庞菲利乌斯方尖碑》和其他著作中，[45] 以及劳尔·皮尼奥里乌斯[46] 在《伊西斯餐桌[...]解释》（这是一部充满埃及象形文字的古代著作，我得知目前在萨伏伊公爵的图书馆中），试图阐明这些象形文字。如果我相信它们能以任何方式有助于理解中国字符，我会建议您获取并使用这些书。中国已故的泰弗诺先生，[47] 国王的图书馆员，曾出版一本关于墨西哥人曾经使用的字符的书。但我相信它们与中国人的或埃及人的都没有任何关系。

关于我刚才在已故安德烈·米勒先生的中国钥匙话题上提到的极尊敬的神父闵明我，我很高兴他友好地记得我，但对他健康不再那么强健感到难过。我恳请您尊贵向他表明我多么尊敬他，以及我多么关心他的健康。我还受托代表极尊敬的神父沃塔[48] 向他致意，他是波兰国王的告解神父，他刚刚离开柏林，他在那里短暂停留，他也曾在汉诺威，在那里他陪伴了普鲁士女王。他现在已离开返回国王，他的恩主，以便在圣周与他在一起。他告诉我，闵明我神父寄给他的旅行记述[49] 去年与国王波兰军队被瑞典国王军队扰乱时其他文件和行李一起丢失了，当时两位国王都在场，波兰人然后是萨克森人也陷入恐慌和恐惧，因为没有什么比恐惧更具传染性。他非常希望获得这份记述，我也一样。因此，如果极尊敬的神父闵明我愿意赐予我们这个恩惠，我请求您尊贵将它寄给我，以及我希望从他和其他朋友那里获得的 天文观察。

您尊贵在您信件最后两页向我传达的，关于至高上帝崇拜的迹象以及古代中国真正启示宗教的痕迹，这些在他们的字符和经典书籍中被发现，在我看来很重要。我一直倾向于相信古代中国人，像古代阿拉伯人（见证约伯记）和或许古代凯尔特人（即[古代]日耳曼人和高卢人），远非偶像崇拜者，而且是至高原则的崇拜者。克卢维里乌斯，无疑是一位杰出的人，在他的《古代日耳曼尼亚》[50] 中甚至相信古代日耳曼人曾对三位一体有一些知识。我完全不同意他的意见。然而，我没有看到任何人因此指责他，而且我发现奇怪的是，对你们这些神父写了古代中国人拥有真正宗教的人造成了如此大的骚动。这有什么害处呢？如果它是错误的，这是一个会产生危险后果的错误吗？[51] 完全不是。[52] [注意划掉的部分："愿上帝让它是真的，而且中国人相信它…… 把情绪放在一边，那[批评]只能来自一些愚蠢的人的偏见，他们充满自己和特殊恩宠…… 他们方便地相信上帝已赐予他们（非常类似于来自吕讷堡地区的年轻女士[53] 她 虽 明智但蔑视世界，因为她相信自己是向她显现的耶稣基督的妻子），而且缺乏…… 足够的慈善，也就是善意，是凯利乌斯·塞昆杜斯·库里奥[54] 的对立面，他写了《论天堂王国的广度……》，而且…… 甚至急于诅咒甚至没有受洗就死去的婴儿。{ 这表明他们没有充分思考上帝的智慧和仁慈。 } 这表明他们对上帝一定有什么奇怪的想法 { 当他们没有充分思考它时}……，如果他们充分考虑他们归于他的后果的话。] 愿上帝让中国人没有其他东西。

至于其余，我恳请您，我极尊敬的神父，记住用不同语言的主祷文。如果能再添加一点最常用词的小目录，那就更好了。同样，请您记住鞑靼地理[即地理记述]，附有对您肯定随身携带的维特森先生地图的评论。同样，请提醒闵明我神父和其他人，为我获取一些古老的中国天文观察；以及几份好的现代观察，尤其是如果您有……年的日食观察，这将有助于牢固确立北京的子午线。[55] 我还想知道是否能更确定地了解所有行星的古代合点，根据它一位古代皇帝开始了中国人用于其年表的周期。[56] 我也重复我关于中国纸张制造的恳求，以及关于将空气封闭在皮革或其他材料中以便即使被相当用力按压，它也完全不通过毛孔泄漏的方法。同样，关于据说从远古以来就在中国的希伯来人，我相信我曾写信告诉您，[57] 以便了解他们是否可能拥有完整旧希伯来圣经，或者像撒玛利亚人那样只有摩西五经和约书亚；[58] 以及他们的文字是什么，是否是撒玛利亚文，这是最古老的，还是犹太人今天仍在使用的、被认为是从巴比伦带来的那种。同样，他们是否有[元音-]点和重音或

没有。[59] 而且如果我向您尊贵或闵明我神父询问了任何其他东西，我重复它并恳请您与您的朋友们在闲暇时考虑它，同样关于施罗基乌斯先生或已故科汉斯基神父的问题。您也不应轻视我相信我提出的关于疾病的问题，其中包括痛风和结石在中国是否像在欧洲一样常见，以及天花和其他天花。 因为所有这些报告都很重要。同样，中国人是否有像奎宁对发烧[60] 和吐根对痢疾一样有效的对某些疾病的补救。我还记得在一些谈论中国的书中读到，如果我没有弄错的话，在马蒂尼神父的地图集中，[61] 中国人有一种在同一年在同一个地方从同一桑树生产两次丝的方法。知道它是否真实以及它由什么组成会很好。

最后，我祈求上帝赐予您持续的健康，并始终赐予您在您美好计划中的成功。怀着热情，

我极尊敬的神父，

您的

附言

我请求您尊贵原谅我借助另一只手，因为我如此忙碌而且不太好。在写完并让人抄写以上内容后，我查看了巴黎《孔子》中的伏羲字符，该书是根据尊敬的神父郭弼恩的记述印刷的，在那里我只找到[组]四个和八个 卦 很好地排列；64 个是混乱的秩序。[62] 但显然添加了推测性的意义，如空气、水等，人们相信或曾相信线性图形表示这些。是否或许您寄给我的印制圆形周围的字符也表示《巴黎孔子》中注明的那些？我请求您检查它，我极尊敬的神父，并让我知道。因为我毫不怀疑您在北京有这本在巴黎印刷的书。

我还必须补充，您尊贵，关于中国字典，自从写这封信以来，我相信我在普鲁士国王的图书馆中找到了戈留斯手稿目录中提到的两本字典之一，即中文和西班牙文的那本。[63] 我看到这本字典，那里是手稿，严格来说解释了中国语言但不解释字符，尽管每个语言词都添加了它的中国字符。以前有人向我谈起一本叫做《海篇》的巨大字符字典，无疑在中

国很容易找到，但在任何欧洲语言中都没有解释。[64] 然而，我在上封信中谈到的另一本字典，或更确切地说词汇，附有事物的插图，[65] 您尊贵让我有希望获得，在我看来是独特且有用的。

如果您尊贵们赐予我寄给我一些东西的恩惠，无论是物理学还是数学、技艺、语言或地理，我请求您尊贵们要么直接寄给我，要么非常清楚地指示您在巴黎的朋友们将其传达给我，否则我将难以利用它。[66] 而且我迄今尚未能够详细了解或收到关于极尊敬的神父丰塔内带回的一切的消息，我再次询问；[67] 或许这些东西已经转到其他人手中。

由于这位神父在他返回后从广州写给我一个字，[68] 而且以非常友好的方式回应了我当他在巴黎时荣幸地写给他的信，我在此通过附上的信回答，我请求您尊贵将它交给他。

我刚才提到的那本中文-西班牙文字典的标题是 Vocabulario de letra china con la explicacion castellana hecho con grand propriedad y abundancia de palabras por el Padre Don Francisco Diaz de la Orden de Predicatores ministro incansable en esto Reyno de China。

关于在鞑靼地区繁殖的骡子，有人向我建议一段来自斯特拉博的文字，说卡帕多西亚的骡子也是如此。[69]

一千多年前在中国的那批基督徒的[刻有铭文的]纪念碑，由基歇尔神父发表的[铭文]——它是否仍然存在？[70] 我想要一份得到中国证据本身充分支持的报告，取自他们那个时代的编年史，以便堵住那些敢于说现代基督徒伪造了它的人的嘴；在我考虑其内容时，这在我看来是不可信的。

以前，我极尊敬的神父，您寄给我一篇用中国鞑靼语的主祷文，但在我看来，您说过是您自己编写的。[71] 因此，如果基督徒鞑靼人使用的那个不同，我请求您将它与其他的一起寄给我。

自从我写了这封信以来，我在普鲁士国王的图书馆中又发现了一本小的拉丁文字典，附有中国字符和词汇，显然是通过尊敬的耶稣会神父们的关注在中国印刷的；但这本小字典只包含最必要的词汇。[72]

最后，我请求您将所附的转交给极尊敬的神父丰塔内。它回答了他 1701 年 9 月 15 日从广州荣幸地写给我的信，我在收到您的信之前几个月就收到了。我正在寄给他一份问题目录，并在请求他方便时考虑它并给我一些见解时，我请求您与他合作，仿佛这也是为了您，实际上确实如此。

附录

我致尊敬的神父丰塔内的信件的摘录：

我极尊敬的神父，

我荣幸地收到您 1701 年 9 月 15 日从广州寄来的信，我非常感谢您对我的记挂，我希望一旦您有机会和闲暇寄给我一些见解，就能再次从中受益。我请求您尊贵，与极尊敬的神父白晋和其他朋友一起，思考所附的问题，除了我现在和以前寄给那位神父的那些[问题]。我祝贺他借助我的二进制算术破译了伏羲线条的谜团。这个发现将对他们那里产生重要后果——在我们向他们提供我们的科学之后不会持续太久。因此我希望，我们将趁我们还有手段时工作，以便欧洲也从中国的科学中获益，比我们迄今为止所做的更多。而且[我希望]我们将努力让皇帝对制作所有技艺的描述产生兴趣，既是为了提供完善它们的手段，也是为了确保中国人不对鞑靼人隐瞒任何东西。这也将是完善皇帝正在制作的字典的方式，因为技艺的术语可以收入其中。[73] 许多人对我不满，既有新教徒，也有法国人，因为我站在你们传教团一边。教皇明智地暂停了对古代和现代中国宗教问题的判断，[74] 因为人们不应谴责未被理解的人。然而，无论他们中一些人的意见如何，重要的是皇帝已宣布自己支持真理。[75] 有人告诉我，自从厄鲁特国王战败以来，[76] 现在从波斯经由乌兹别克，

以及从莫卧儿国都可以去中国。但如果人们能像您那样通过海路同样容易地去，那么那条路线将优于所有其他路线。

尾注

1. 莱布尼茨正在回应信件 I，他尚未收到白晋 1702 年的信（K）。

2. 莱布尼茨在此进一步详细说明了他对与中国进行单方面智力交流的持续担忧。参见信件 B，注 31。

3. 弗朗西斯·梅库里乌斯·范·赫尔蒙特男爵（1618-1698），著名医生和炼金术士扬·巴普蒂斯塔·范·赫尔蒙特（1579-1644）的儿子，在他生命的最后几年成为莱布尼茨的亲密熟人，当时他是汉诺威宫廷的客人。莱布尼茨甚至为他用拉丁文代笔写了一本关于创世记前四章的书。他是一位训练有素的炼金术士和卡巴拉学者，在许多领域知识渊博，包括绘画。莱布尼茨在许多领域——实践和理论——向他寻求建议。关于这位迷人人物的进一步细节，见 A. P. 库德特，《十七世纪卡巴拉的影响：弗朗西斯·梅库里·范·赫尔蒙特（1618-1699）的生平和思想》（莱顿：布里尔，1999）。关于范·赫尔蒙特与莱布尼茨的关系的细节，见同上，“莱布尼茨与卡巴拉”，第 309-329 页。关于莱布尼茨对卡巴拉的详细研究，见 A. P. 库德特，《莱布尼茨与卡巴拉》（多德雷赫特：克鲁威尔，1995）。

4. 西班牙王位继承战争（1701-1714）是由西班牙国王查理二世的去世引发的，他没有子嗣。由英格兰和奥地利领导的大同盟成立，以挑战法国对西班牙王位的权利主张。普鲁士和其他德意志邦国站在大同盟一边反对法国。

5. 关于北方大战，见信件 H，注 14；17。

6. 诺特堡（今什利谢尔堡）位于拉多加湖畔的涅瓦河上。在 1703 年圣彼得堡建立（作为彼得大帝早期对瑞典胜利的象征）之前，它是英格尔曼兰/英格里亚的主要城市，该地区位于芬兰湾以南和以东。

7. 奥斯曼帝国的军队在 1683 年维也纳城门被奥地利人（在波兰人的帮助下）击败，标志着“大土耳其战争”（1683-1699）的开始。战争期间土耳其人的进一步失败导致大部分匈牙利落入奥地利人之手。苏丹苏莱曼二世于 1691 年去世，因此莱布尼茨指的是最近过去的事件（正如他经常在向白晋发送欧洲新闻时所做的那样），而不是当前事件。在 1703 年初写这封信时，穆斯塔法二世自 1695 年起一直是统治苏丹，莱布尼茨显然知道这一点。

8. 鞑靼人，他们自称“鞑靼人”，是突厥语系的穆斯林。在几个世纪中，克里米亚半岛和俄罗斯南部的鞑靼人逐渐被置于沙皇的统治之下。

9. 莱布尼茨已在信件 H，注 10 中讨论过这个解决方案。

10. 莱布尼茨在信件 B，注 33-35 中简要提及他与笛卡尔派在需要和力的性质上解释物质和运动以用于新力学科学的意见分歧。“实在形而上学的崇高原则”指的是需要将亚里士多德的实体形式概念重新引入力学作为力的概念。用莱布尼茨的话说，“……在试图考察力学的根本原则以解释我们从经验中学习的自然法则时，我意识到仅考虑 扩展质量是不够的，而且有必要此外，使用 力的概念，它是非常可理解的，尽管它属于形而上学的领域。”AG 139；G IV 478。莱布尼茨也在信件 H，注 31–33 中相当详细地讨论了他的动力学理论。

11. 莱布尼茨重复了他在信件 H，注 28–29 中的评论，但添加了法国科学院物理学家和成员埃德梅·马里奥特（1620?–1684）的名字。马里奥特，圣马丁-苏-博恩的院长，靠近第戎，曾写过一本关于笛卡尔力学的著作，《论物体的碰撞，或物体撞击，其中主要运动规则由其真正原因解释和证明……》（第 3 版：巴黎，E. 米沙莱，1679）。

12. 莱布尼茨确实对“我的二进制算术与伏羲模型的关系”如此“惊讶”，以至于甚至在他在这封信中回应白晋的发现之前，他已经在几封通信中详细讨论过它。莱布尼茨专门写了一整封信给沃塔神父（关于沃塔，见本信第 29 页及下文注 48），在结尾为“或许不合您口味”的事情道歉。（Z 291）几天后（1703 年 4 月 7 日），他将关于这个“中国联系”的一节添加到他向法国科学院提交的关于二进制算术的就职演讲中，在给其主任让·比尼翁神父的随附信中强调了这一添加，并建议用它替换他早期的论文。（Z 292-301）可以理解的是，莱布尼茨在给沃塔神父的信中讨论了二进制算术对中国异教徒证明创造教义的重要性，但在给法国科学院的论文中省略了它。

13. 注意。见艾伦的黄色笔记页关于这一点。

14. 波斯波利斯是波斯古都，现在在伊朗南部，离设拉子不远。切希尔米纳尔（意为“四十宣礼塔”）是莱布尼茨时代波斯人对波斯波利斯的称呼。后来在他《神正论》（1710）中关于琐罗亚斯德和古代波斯人宗教的长篇讨论（§137）中，莱布尼茨特别提到他们的仪式在“波斯波利斯或切尔米纳尔（意为四十柱）的辉煌废墟”中的雕刻表现。

15. 莱布尼茨在此指的是古波斯楔形文字。在波斯波利斯，它通常以三种语言形式出现，伴随埃兰语和阿卡德语（也用楔形文字书写）。文字的破译是一项集体努力，历时多年（1798-1932）。二十世纪的学术意见同意莱布尼茨的观点，即这种书写系统可能确实早于大流士一世和薛西斯的君主制。见 R. G. 肯特，《古波斯语：语法、文本、词典》（第 2 版修订；纽黑文：美国东方学会，1953），第 9-12 页。

16. 唐·加西亚·德·席尔瓦·伊·菲格罗亚（1574-1628）在 1618-1619 年他的外交任期内访问了该遗址。他的日记被翻译为《唐·加西亚·德·席尔瓦·伊·菲格罗亚在波斯的使节》，由 F. 德·维克福特（巴黎，1667）。（我们感谢加州大学伯克利分校的大卫·斯特罗纳赫提供本注及前一注中关于波斯的资料和参考。）莱布尼茨还提到一位荷兰大使曾以高昂代价让画家绘制废墟草图；见《神正论》§137。

17. 梅尔基塞德克·泰弗诺（1620-1692），欧洲旅行家和语言学者，直到去世前一直与莱布尼茨通信。从 1684 年起，他是巴黎的皇家图书馆员。他将许多其他旅行家的记述汇编编辑成两卷本著作，《各种奇特旅行关系，这些旅行未曾出版，并且已从法国、西班牙、德国、葡萄牙、荷兰、波斯、阿拉伯……旅行者的原件翻译或摘录》。巴黎：托马斯·莫埃特，1696。关于波斯的材料题为“波斯波利斯古迹描述”（从英文翻译）。

18. 白晋进一步阐述了莱布尼茨在信件 I 中提出的二进制与创造的类比，应用他自己关于中国字符二进制起源的理论来证明一种通用特征的可能性，该特征能够表达创造的性质以及所有受造物。见信件 I，passim。关于莱布尼茨对通用特征的看法，见信件 H，注 40。

19. 莱布尼茨重复他在信件 H 中简要概述的（新）柏拉图主义创造理论，该理论将所有受造物的限制等同于缺陷或否定。见注 19；22。

20. 莱布尼茨实际上阐明了白晋的中国策略：如果基督教能够向中国人证明他们最古老的字符具有他们已丢失的原始意义，但被基督教西方恢复，那么通往他们皈依的道路将大大 облегчена，因为皈依基督教，他们将恢复他们自己的根源，而不是放弃它们接受外来宗教。莱布尼茨明确将这种用于皈依中国人的策略与对犹太人所做的相同进行比较：即，“能够比[中国]博学者自己更好地解释他们的古代书籍。谁不知道在我们今天，基督教学者比犹太人自己更好地解释希伯来人最古老的书籍？”见他的“论孔子的公民崇拜”，载 CR 64；WB 252-254。莱布尼茨自己作为新教徒使用这种策略反对天主教徒，证明后者不忠于教父的文本；见巴鲁齐，第 373 页。

21. 见 1701 年 11 月的 Demonstratio quod columnae serierum . . . sint periodicae，GM VII: 235-238。关于莱布尼茨使用二进制记法的各种细节和参考，见 Z 178-181 et passim。

22. 鲁道夫·范·屈伦（1540-1610），莱顿大学数学和军事科学教授，曾将 π 的值（也称为鲁道夫数或鲁道夫数）计算到 35 位小数。尽管现在借助超级计算机已将 π 计算到超过二十亿位，但尚未找到确定这些计算的代数公式。关于屈伦，见《科学传记词典》，编者 C.C. 吉利斯皮（纽约：斯克里布纳，1981），III 181a-b。

23. 莱布尼茨在此指的是白晋在信件 I 中努力发展某种基于二分法或二元区分的通用特征。（上文注 18）。白晋发现古代中国《易经》字符的二进制基础帮助复兴了莱布尼茨毕生努力发展适当字符以用于通用特征，或如他常称的“人类思想字母”。它也可能是西方一直在寻找的作为轻松和系统地破译中国字符手段的《中国钥匙》（Clavis Sinica）的基础。在 1694 年安德烈亚斯·米勒去世后，对这样一把“钥匙”的希望破灭了（见信件 B，注 8）。如果这些古代字符是中国表意文字的基础，这将意味着中国语言具有算术基础，可以作为通用特征或“思想演算”的模型。见袁婷·赖，“莱布尼茨对中国和永恒哲学的研究”，il cannocchiale 10（1999 年 1-4 月），第 127 页。

24. 莱布尼茨指的是他最早的作品，他的 1666 年《组合技艺论文》A VI: N. 8；选段见 L 73-84）。莱布尼茨在信件 B，注 36 中被删去的一段中出色地描述了他的目标。莱布尼茨在此重申了他理想“思想演算”的主要科学和哲学目的（除了可能作为容易学习和可交流的通用语言）：即，作为一种判断方法，可以检查论证的有效性，以及作为一种发明方法，可以发现新真理和论证。

25. 正如莱布尼茨几年后所说：“伏羲，中国最古老的君主和哲学家，已理解万物从一和无的起源，即，他的神秘图形揭示了与创造的某种类比，包含二进制算术（并且还暗示更伟大的事物），我在几千年后重新发现它，在那里所有数字仅用两个记号 0 和 1 书写。”“关于中国礼仪和宗教的评论”（1708），CR 373；GP II 383。所暗示的“更伟大的事物”可能是莱布尼茨自青年时代以来一直寻求的通用特征。

26. 莱布尼茨在此指的是《易经》64 卦的传统解释，它被视为一本占卜书和智慧书。关于背景以及每个卦的解释，见 R. 威廉的英文译本《易经或变化之书》，C.F. 贝恩斯译

（第 3 版；普林斯顿：普林斯顿大学出版社，1967）。关于如何咨询《易经》的描述，见 CR 20-22。

27. 莱布尼茨在他的《形而上学谈》（后来被删去的一段）中简洁地总结了这种传统的认识论区分："当我仅通过经验知道一物的可能性时，我对它的观念/知识是混乱的；因为一切存在的事物都是可能的。因此我们这样知道物体及其性质。但当我能先验地证明可能性时，这种知识是清晰的。"§XXIV（AA VI iv，第 1567 页，注 l. 18；LG 41，注 a）

莱布尼茨在此批评白晋在他上一封信中概述的特征努力，因为他依赖未经分析的经验区分（例如颜色），而通用特征（UC）或物种的字符必须是分析上透明的，可以这么说。它们必须捕捉所指对象的特定属性，以便我们能够分析它，从而能够将其还原为其基本元素，并在其与其他事物组合时检测其独特性质。因此莱布尼茨继续（这一次在《形而上学谈》的实际文本中）以一个清晰知识的例子："……这就是鉴定师的知识，他通过某些测试或标记来辨别真金与假金，这些测试或标记构成了金的定义。"同上，第 1568 页；LG 41。莱布尼茨的化学例子很贴切：化学表和公式是 UC 基础的极好例子，因为它们允许我们检测一个元素的特定属性（单独或在化合物中）以及发现其其他可能的化合物，因为其电子的数量和位置等。

28. 这些"新实验"将使我们能够将一切还原为其自然的"元素"，因此正如我们可以在化学中确定可能的和不可能的化合物一样，我们也可以用适当的 UC，使用逻辑和物理的演绎法则来检测虚假论证和错误实验。以数学家解决论证的方式，科学家和哲学家同样可以"坐下来拿着他们的笔和算盘……并互相说'让我们计算吧！'（calculemus！）。"A VI iv，第 193 页；以及 A I iv，第 315 页。

29. 早在 1666 年，艾萨克·牛顿（1642-1727）就使用玻璃棱镜将一束阳光扩散成光谱的颜色。虽然牛顿关于颜色性质的重大发现在 1704 年出版的他的巨著《光学》中被纳入，但他早得多地在讲座和信件中公开传达了他的发现，通常是给伦敦皇家学会的。后者，写给学会秘书亨利·奥尔登堡，在学会的《哲学汇刊》上发表。在信件 N 中，莱布尼

茨说牛顿的工作“完全颠覆了笛卡尔给我们的关于光线性质的想法。”关于笛卡尔理论，见信件 N，注 17。

在牛顿之前，黑和白被视为光谱的两个质上不同的极，每个“颜色”在它们之间仅由真颜色黑和白的某种渐变或混合构成。牛顿的实验证明黑不是一种独特的颜色，而是所有颜色的缺失；白，以前被认为是颜色中最纯或最亮的，无非是所有其他颜色的混合！因此黑和白根本不是真颜色，而那些“在之间”的在其数学上可测量的渐变中实际上是真正的原色。通过否定黑和白的极性对立——像光和暗、热和冷等的极性对立，即任何质的区分——牛顿有效地摧毁了经院亚里士多德传统中作为科学解释形式的基本性质的度和亲和性。关于牛顿贡献的精彩讨论以及当时适当书面来源的丰富参考，见 P.S. 韦斯特法尔，“牛顿颜色理论的发展”，《伊希斯》，LIII（1962）：339-358。

30. “例子不需要真理”。H. 扎赫尔提到 1703 年 4 月 L. 库图拉（C 327）中的一个片段，其中莱布尼茨用这个短语指白晋的例子。（见 Z 284）

31. 根据亚里士多德理论——如经院学者所理解的——物理变化的发生是因为四个基本元素中的每一个，土、气、火和水，可以通过四个基本性质中的两个的组合在某种程度上被限定或改变，即热、冷、湿和干。

32. 理查德·斯温斯海德或苏塞特，被称为计算者，在 14 世纪中期活跃。他最著名的是他试图将数学和演绎技术应用于计算物理变化。在他最著名的论文的第一章《计算者》……（威尼斯 1520）中，使用传统的亚里士多德定性变化解释（前注中提到的），他讨论了特定性质的强度（或缺乏：即缓和）的概念。见 L. 桑代克《魔法和实验科学史》，第 3 卷（纽约：哥伦比亚大学出版社，1934）第 XXIII 章，“计算者和数学物理学的兴起”。

莱布尼茨赞赏斯温斯海德在物理学中量化问题的努力——尽管他认为它们不成功——而不是沉迷于传统的经院辩论方法。见 CO 177；340。

33. 莱布尼茨重复了他的主张，即二进制算术的基础（0 和 1 单独以及各种组合）是创造的进一步证据。（见信件 H，注 19。）除了验证他自己的发现，伏羲六十四卦的二进制基础证明了古代中国思想理解创造教义，在圣经启示之前或之外。这种智慧对莱布尼茨来说证明了古代中国人的逻辑敏锐性和智力，这些品质将提高欧洲对中国思想和文明的尊重。它还将使传教士能够声称，通过皈依基督教，中国人只是返回他们自己古代的来源和传统。因此，例如耶稣会士可以在他们的传教努力中适应这些传统。关于耶稣会士的适应主义立场，见 D. 蒙格罗，《好奇的土地：……》，第八、九和十章。

34. 数字 7 在圣经传统中被认为是神奇的或神圣的，因为创世记（II: 2-3）中的创造记述："到第七日，神造物的工已经完毕；……神赐福第七日，定为圣日。"二进制算术证实了这个观念，因为数字七（或七进制）是用三个 1 书写——这是三位一体的暗示。虽然莱布尼茨注意到二进制算术数字象征中对三位一体教义的这种"证实"，但作为理性主义者和数学家，他是否非常相信所有这些数字神秘学是值得怀疑的。他最有可能自愿提供这样的"洞见"，因为他认为这种方法可能增加基督教在中国人眼中的尊重。

35. 米勒编辑了马可·波罗东方旅行记述的拉丁文版，1671 年。关于米勒的更多细节和参考书目，见信件 B，注 8。

36. 关于闵明我，见信件 B，注 7。

37. 米勒曾负责勃兰登堡选帝侯弗里德里希·威廉（1640-1688）——被称为"大选帝侯"——在柏林的东方收藏。关于神圣罗马帝国中选帝侯的角色，见信件 H，注 1 的注 1。

38. 关于戈留斯，见信件 H，注 52。

39. 莱布尼茨在本信的附言中给出了这部作品的完整标题，他可能在 1703 年初从柏林返回汉诺威时添加了它。

40. （注在所提到的目录的上下文中继续。）

41. 通过否定埃及象形文字与中国字符之间的任何联系，莱布尼茨有效地削弱了古代神学支持者的一个关键论点。见 D. 库克 & H. 罗森蒙特，“莱布尼茨、白晋、古代神学教义和《易经》，《关于中国的最新消息》，第 135 页。最著名的埃及-中国联系支持者是阿塔纳修斯·基歇尔；关于基歇尔，见信件 E，注 11-12。

42. Horapollo= Horus Apollo=Orus 或 Orion。Orus 是来自埃及底比斯的希腊语法学家和修辞学家，生活在公元 5 世纪。他是著名新柏拉图主义者普罗克洛斯的老师，并著有一部部分现存的关于希腊词源学的著作。

43. 阿米安·马尔切利努斯，一位希腊血统的罗马历史学家，生活在公元 4 世纪。作为士兵，他游历了整个罗马帝国、小亚细亚和埃及。他写了一部名为《历史三十一书》的著作。有一本 1474 年罗马版的第 XIV-XXVI 书，由保卢斯·萨比努斯编辑（前 13 本书已不存）。后来的 1533 年版（在奥格斯堡和巴塞尔）添加了第 XXVII-XXXI 书。在这本书的第 XVII 章中，马尔切利努斯提供了罗马竞技场中竖立的一根埃及方尖碑上象形字符的解释（§6）。在本章稍后（§§18-23），给出了这根方尖碑上一些象形文字的详细希腊语翻译。

44. Pierus Valerianus=Giampetro Valeriano（1477-1558），意大利学者，其赞助人包括教皇尤利乌斯二世、利奥十世和美第奇家族。他写了《埃及圣物象形文字……》（巴塞尔 1566[仅第一部分]；法兰克福，1678）。

45. 关于基歇尔，见信件 E，注 11-12。

46. Laurentius Pignorius=Lorenzo Pignoria（1571-1631），意大利学者和古物学家，写了一部关于埃及象形文字的重要著作，曾多次再版并改名。莱布尼茨以其第三个标题引用它：《伊西斯餐桌，关于埃及人的圣物……》（阿姆斯特丹：A. 弗里西，1669）。

47. 关于泰弗诺，见上文注 17。在上述他的《各种奇特……》中，有一篇题为《托马斯·盖奇的墨西哥和新西班牙关系》的选段，已单独出版（巴黎：T. 莫埃特，1696）。它是一本对开本，有标题页和 58 页。前 46 页是象形文字，盖奇在最后十几页中仔细解释。这些后几页单独题为《用墨西哥语解释的墨西哥历史图，然后用西班牙语》。

48. 卡洛·毛里齐奥·沃塔神父，S.J.（1629-1715）是教皇驻华沙使节和波兰国王的告解神父。莱布尼茨在此期间（1698-1709）与他通信，特别是关于俄罗斯和中国的事务。沃塔从 1684 年到 1689 年一直是莫斯科耶稣会院的负责人。

49. 这可能是他 1691 年试图通过另一条陆路返回中国的失败尝试的记述，因为“莫斯科关闭对他”，闵明我试图“从波斯通过乌兹别克鞑靼人和布哈拉的土地前往中国”，但发现太危险。《最新中国》，前言，A IV vi，第 406 页；CR 57。另见下文注 76。

50. 《古代日耳曼尼亚三书》，由菲利普·克卢维里乌斯=菲利普·克吕弗（1580-1623），德国地理学家和历史学家，在 17 世纪初在莱顿多次再版，他在那里活跃。

51. 莱布尼茨不愿接受白晋的理论，即“古代中国人拥有真正宗教”，但尽管如此，在较弱的、实用主义的基础上认识到他的方法的价值。见 D. 库克 & H. 罗森蒙特，Jr.，莱布尼茨、白晋、古代神学教义和《易经》，《关于中国的最新消息》，第 135-136 页。

52. 人们可以看到为什么莱布尼茨可能会删除这段——不想冒犯他的通信者的天主教情感[关于诅咒异教婴儿和相信教会之外无救赎]，但它很有启发性。

53. 莱布尼茨在此指的是罗莎蒙德·冯·德·阿塞堡（1672-1727），她在 1691 年秋天（与她的姐妹们一起）出现在汉诺威宫廷，留下强烈印象。莱布尼茨对她的主张非常怀疑，并用自然主义、心理学术语解释她的行为。尽管如此，他说应该让她独自一人（他将她比作罕见的“橱柜件”），基于实用主义理由，因为她的活动和异象有积极结果。（A I vii: 34-35）。关于莱布尼茨对这类狂热者（Schwärmer）的态度，见 D. J. 库克，“莱布尼茨论狂热”，载《莱布尼茨、宗教与神秘主义》，编者 A. 库德特、R. 波普金 & G. 魏纳（多德雷赫特：克鲁威尔，1998），第 107-135 页。关于阿塞堡的特别讨论，见同上，第 121-123 页。只要这类神秘主义者不破坏或扰乱公共秩序，莱布尼茨的态度就相当开放。关于她最主要追随者的英文译本，见 J. W. 彼得森，《致一些神学家的一封信……附有上帝赐予一位高贵少女的确切记述》（伦敦：J. 惠特洛克，1695）。

54. 凯利乌斯·塞昆杜斯·库里奥（1503-1569），意大利作家和语言学家，写了《论天堂王国广度的对话……》（巴塞尔 1554；法兰克福 1617），主张上帝救赎的丰满。他相信遵守自然法准则的异教徒将得救。在他生命的早期，他捍卫路德宗和改革宗教义，并最终在瑞士找到庇护，作为巴塞尔大学文学教授免于宗教裁判所。

55. 显然莱布尼茨忘记了他所指的日食日期，并计划稍后插入它。绘制子午线（环绕地球并通过两极和它们之间任何给定点的线）是 17 和 18 世纪持续的科学关注问题。关于通过巴黎绘制本初子午线的努力，见莱布尼茨给白晋的第一封信，注 29。通过本初子午线（例如格林威治）绘制经度线（用地球 360°圆周的度数或 24 小时时钟的分数表示）对于适当导航和制图至关重要。关于寻找适当经度的迷人搜索，见达瓦·索贝尔的书《经度》。

56. 莱布尼茨指的是传说中的统治者颛顼（公元前 2513-2435 年）。见 WB 738，注 52。

57. 直到这封信，莱布尼茨从未直接向白晋询问关于中国犹太人的事，但他早些时候向韦尔瑞斯和郭弼恩询问过他们的圣经。莱布尼茨以及他的耶稣会通信者对中国犹太人的

存在感兴趣有几个原因，后来他告诉白晋“彻底考察他们的教义和礼仪很重要。”（信件N）首先且最重要的，莱布尼茨（和基督教欧洲）渴望验证旧约文本，因为它的预言是世界摩西年表的主要来源。莱布尼茨，像许多其他人一样，致力于将此类记述与当代天文和地质发现调和。因此，“至少复制他们的创世记开头很重要，以查看他们的族长家谱是否或许与七十士译本文本一致。”（给韦尔瑞斯的信，1700 年 1 月 1 日[A I xviii，第 273 页]）其次，作为正统基督教的捍卫者，莱布尼茨希望前基督教时代的更早希伯来旧约文本可能包含“欧洲犹太人可能出于对基督教的反感而更改或删除的书籍或段落。”（同上。）最后，如果犹太人“从远古以来”就在中国，这将增加对白晋和其他人的“形象主义”理论的可信度：即这些犹太人是诺亚、族长或“失落的十支派”的后裔，因此是中国古代经典文本中嵌入的神圣真理的来源。

在信件 O（未回复）中，他给白晋的最后一封信（见注 2），莱布尼茨转发了柏林宫廷讲道士新教神学家丹尼尔·雅布隆斯基（1660-1741）关于中国犹太人的一系列问题，他编辑了一版旧约（1699）。大约在同一时间，莱布尼茨摘录了一些关于中国犹太人的材料（见 WB，第 590-593 页）。关于中国犹太人的文献非常丰富，尤其是开封的古老社区。最全面的作品是 D. D. 莱斯利，《中国犹太人的生存：开封犹太社区》（莱顿：布里尔，1972）。最新的努力是《从开封……到上海：中国的犹太人》，编者 R. 马莱克（内特塔尔：施泰勒出版社，2000。特别见 R. 维达迈尔，“关于 G.W.莱布尼茨通信中中国犹太人问题的讨论”，同上，第 559-588 页。

58. 撒玛利亚人传统上被认为是当地人口和巴比伦流放后留在北部以色列（即撒玛利亚）的犹太人的后裔。撒玛利亚摩西五经与犹太-基督教摩西五经或摩西五书有细微但重要的差异，是他们基本教义、仪式和传统的唯一来源。撒玛利亚《约书亚记》（作为后来材料的汇编而存在，仅用阿拉伯语）不应与——正如莱布尼茨显然误以为的——希伯来《约书亚记》，即六经的第六卷混淆。撒玛利亚摩西五经的希伯来文本仅在 17 世纪初才到达西欧。它在圣经学者中引起了巨大兴奋，他们认为（结果证明不正确）它可能是比犹太人（拉比）使用的传统马所拉文本更古老的希伯来圣经版本。《犹太百科全书》，XIV，栏 265-266。

59. 用于转录和传递摩西五经的希伯来文字（即“方形”或“亚述”文字）确实是由巴比伦的犹太人采用的，并取代了古希伯来文字。“撒玛利亚字母表是早期希伯来文字的唯一后代，至今仍在使用。”同上，栏 267。撒玛利亚人对摩西五经希伯来文本的元音和阅读重音系统与马所拉系统完全不同。同上，XIII，栏 1468。

60. 见 WB 739，注 55-56。

61. 关于马蒂尼，见信件 H，注 47。莱布尼茨在此指的是马蒂尼的主要制图作品，《中国新地图集》（阿姆斯特丹：J. 布劳，1655）。

62. 关于郭弼恩和他的《孔子中国哲学家》……，见信件 E，注 22。郭弼恩给出了所谓的文王（也称为后天）《易经》卦序，而白晋提供了所谓的“伏羲”序（也称为先天序）。

63. 关于戈留斯，见信件 H，注 52。斯坦达尔特提到“他的图书馆销售目录中三件物品，该目录于 1696 年公开出售”：……“一本用西班牙语解释 10000 个字符的‘宝库’；……一本由戈留斯注释的西班牙-中文词汇；以及一本包含中国字符发音的西班牙小册子。”N. 斯坦达尔特，《基督教在中国手册》，第 871 页。

64. 约布·卢道夫，著名的德国语言学者，1698 年写信给莱布尼茨关于《海篇》。但莱布尼茨以前通过他转发给白晋的科汉斯基神父的问题已经听说过这部作品（A I xiv: N. 450，问题第 22 号）。关于欧洲《海篇》版本的混淆，见 D. 蒙格罗，《好奇的土地……》，第 217-218 页，以及 WB，第 649-50，注 16）。关于《海篇》这个名称的确切字典一直有一些混淆，因为在中国没有单一一本叫这个名字的字典；相反，明代许多字典的标题中都有“海篇”这个词，以至于后来这些被视为一个类别本身。这些在很大程度上直接源自金代和明初的几部早期作品，其标题包含“篇海”这个词。经史海篇直音，最常简称“海篇”，特别通过张黼（约 1544 年活跃？）的重新发行并略有修改的版本传播，该版本题为新校……（新校……），从至少 1544 年开始多次印刷；其中几个版本仍然存在。这

个“新校”版本非常庞大，包含 55665 个单独的字符条目，其中许多是简单的图形变体，但这部作品的更大重要性在于它是自汉代以来最早的按一组重复出现的组成图形元素分组组织字符的词典之一，而不是按发音组织，作为直接影响 214 个标题（“部首”或语义限定符）系统的发展的词典格式的模型，该系统由梅膺祚（1570-1615）在他的巨著《字汇》（约 1615；字符汇编）中正式化，其中包含 33179 个字符。这是 1716 年康熙字典（康熙字典）包含 47035 个字符的模型，该字典在 214 个图形分类标题下。见远藤由里子，“海篇诸本考”，《梅光女学院大学论集》22（1989）：31–57；大岩本幸次，“明代「海篇类」字书群に関する二三の問題”，《东北大学中国语学文学论集》4（1999）：19-33；以及杨正业“海篇成书年代考”，《辞书研究》（2005.1）：174-176。

65. 见信件 H，注 52。

66. 莱布尼茨已经怀疑他没有收到白晋通过巴黎寄给他的一切。他的怀疑很快得到证实，因为莱布尼茨再也没有收到白晋的答复，包括他 1704 年 9 月 15 日写给比尼翁神父的长信，该信本应也转发给莱布尼茨（见 C. 冯·科拉尼，《中国科学学院……》，第 26-27 页）。

67. 莱布尼茨在本信结尾给丰塔内加了一个注。正如他下面的评论所明显的那样，莱布尼茨以为丰塔内仍在中国的，不知道他已在 1702 年 12 月乘坐“安菲特里特号”离开中国。他于 1703 年底乘坐一艘英国船抵达欧洲，并最终于 1704 年 6 月 13 日从巴黎回复莱布尼茨（WB: 436-439）。

68. 见他 1701 年 9 月 15 日给莱布尼茨的信。（WB 446-451）。

69. 斯特拉博（公元前 64 年？—公元 24 年？）是一位著名的希腊地理学家和历史学家。在他的著作中找不到这样的段落。通信的德国编辑认为莱布尼茨很可能指的是小普林尼的《自然史》，8. 173（WB 740，注 64）。

70. 著名的碑刻于 1623/25 年在西安附近被发现（现存于西安碑林博物馆），证明景教或亚述基督徒在唐代（618-907 年）在中国活跃。碑上的铭文日期为 781 年，“由 1800 个字符组成，并分为几个部分。”N. 斯坦达尔特，《基督教在中国手册》，第 3 页。来自碑刻的中文文本呈现了景教徒的神学、历史和祈祷，并在基歇尔的 1667 年《中国图说》中复制。关于碑刻的最新信息以及关于其真实性、何处何时被发现等的争议，见同上，第 3-4 页；12-15 页 et passim。关于“景教碑”以及它在基歇尔中出现的文本的详细讨论以及优秀复制，见 D. 蒙格罗，《好奇的土地……》，第 164-172 页。

71. 由白晋与信件 E 一起寄给莱布尼茨。

72. 这样的卷本已不再能在柏林国家图书馆找到（WB 740. 注 67）。

73. 白晋在信件 F，第 5 页中提到了这个满汉字典的项目。

74. 在写这封信时，教皇尚未就传统儒家宗教是否应被视为与基督教教义不相容的无神论或偶像崇拜形式发布教皇诏书。莱布尼茨一再站在耶稣会一边，他们认为基督教可以理论上和实践上适应这样的中国信仰以及他们持续的祭祀仪式。见信件 B，注 1，关于所谓的“礼仪之争”。教会最终在接下来的几十年中站在莱布尼茨和耶稣会的反对者一边，发布了教皇诏书 Ex illa die（1715）和 Ex quo singulari（1742），这些诏书否定了他们对中国礼仪的解释。见 D. 蒙格罗，《好奇的土地……》，第 357 页。莱布尼茨在生命的最后一年（1716）写的最强烈和最长的努力捍卫适应主义观点的是他的《中国自然神学论》。在《论》中，莱布尼茨使用从尼古拉斯·雷蒙德在 1715 年秋天收到的朗戈巴迪和圣玛丽的反适应主义论文中的完全相同的中国材料——来证明他相反的观点。关于相关通信的细节和《论》的批判版，见 LP。关于《论》的英文译本和研究，见 CR。

75. 通过 1692 年颁布宽容敕令，允许在中国实践基督教，康熙皇帝实际上站在耶稣会一边，尽管他仅在中立意义上肯定基督教可以被容忍，因为它与中国的宗教不相容。

1700 年，皇帝更积极地回应，通过批准某些特定的耶稣会立场。“北京的耶稣会士曾征求康熙皇帝对他们‘适应’基督教的三个主要问题的意见（天指至高存在；祖先崇拜是感恩的表达；孔子崇拜是公民非迷信仪式），皇帝肯定这确实是正确的解释。”N. 斯坦达尔特，《基督教在中国手册……》，第 498 页。郭弼恩曾在 1701 年 11 月 10 日从巴黎写信给莱布尼茨传达了这次成功向皇帝请愿的细节（WB 378-381）。后来这些皇帝意见的文本以及其他文件，被题为（足够清晰地）Brevis Relatio eorum quae spectant ad declarationem Sinarum Imperatoris Kam Hi circa coeli, Cumfucii et avorum cultum, datum anno 1700（奥格斯堡和迪林根：本卡德，1703），被送往罗马。这次努力被拒绝，因为罗马对一个异教徒敢于对基督教信仰和教义事务发表意见感到冒犯。（见 WB 729 和 C. 冯·科拉尼，《中国科学学院》，第 13 页）。

76. 厄鲁特，更为人所知的是卫拉特，指的是现在被称为外蒙古阿尔泰地区的地区。当时由卫拉特或西部蒙古人居住，包括中国新疆（准噶尔）和西藏东北部地区。从 17 世纪晚期开始，卫拉特或西部蒙古人逐渐被满人控制，到 18 世纪末，满人主导了整个外蒙古。见 O. 拉铁摩尔，《中国内亚边疆》（波士顿：灯塔出版社，1967）第 78-79 页；86-87 页。自从通过俄罗斯的陆路对中国关闭给耶稣会士（见信件 H，注 15），欧洲希望通过波斯或北印度（莫卧儿帝国），通过土耳其斯坦（“乌兹别克”）找到一条替代路线到中国。然而，正如莱布尼茨在此怀疑的，这样一条路线从未实现。

## 18 May 1703 Leibniz to Bouvet

Translation and Annotations by Alan Berkowitz and Daniel J. Cook

My Very Reverend Father,

Your Reverence's letter dated Peking, the 4th November 1701 was delivered to me the 1st of April of the year 1703 by the care of the Very Reverend Father Gobien. I am delighted to learn that Your Reverence is well and that the affairs of the Europeans go well in China, since it concerns the progress of the Christian religion.[1] But I cannot exempt myself from urging you and your companions to take advantage of the favorable times a little more than you have up until now, in order for us to obtain in exchange of our own the fine knowledge of the Chinese, especially in the arts, since a passage of your letter makes me believe that the good will that they have there for foreigners will grow extremely cold one day when they believe that they no longer have so much need of them. This passage is the one where you relate, my Reverend Father, that the emperor for some years now, having attained to his satisfaction what he desired to know of the theory of our sciences, no longer gives the same opportunity to the Europeans to converse with him on these matters. Being as such, it is extremely important to work on the history of the arts of China without delay while we still have the means, and more so since it is easier for them to learn our sciences and knowledge than for us to learn theirs, as much because ours consist more in reasoning and theirs more in experience, as because ours are public for the most part but those of the Chinese are scarcely known except by professionals, and are passed on there as if by way of tradition [i.e., private, oral transmission].[2] I believe that you could artfully interest the emperor himself in this research of the arts, making him realize that the way of perfecting the arts of his country is to give them descriptions; and by this method one will be able to see what the Chinese lack in order to make up the deficiency with the help of the Europeans. And likewise, that it is important to have a complete dictionary, or rather several which make up a complete one, which contains all of the characters especially particular to each profession, and that it is important to the Tartars that the Chinese not hide anything from them. Now, on the subject of terms, when it come to explaining them there should be given illustrations of natural and artificial things and descriptions of methods or secrets. In particular, my Reverend Father, I ask you to obtain for me a description as clear as possible concerning the manufacture of all kinds of Chinese paper and the methods of embellishing it. From what are the types [i.e., printing fonts] made which they use to print their characters? Formerly, Monsieur van Helmont the younger[3] was telling me that they made them from a lacquer or varnish in which they easily engraved these characters while it was still soft, but that afterwards it hardened like wood; that would be convenient. I would also hope to learn whether they have good methods of strengthening leather or fabrics in such a way that air and water cannot at all penetrate, and especially air, because I have not yet found to my satisfaction that anyone has succeeded in this in Europe in order to make large bags or cushions that could be inflated and slept upon.

As to the affairs of Europe, they are in such a state that they make us envy the Chinese. The death of the king of Spain gave birth to a terrible war which will decide the fate of Europe.[4] The war between the czar and the king of Sweden still continues.[5] But as the latter [i.e., the king of Sweden], instead of pushing back the Muscovites, turned his forces against Poland and entered into the heart of this kingdom, the Muscovites regained vigor and are greatly harassing the provinces of Sweden which are opposite them, even taking Nöteburg, a place of some importance[6] They also are trying to spread out along the side of the Black Sea, building a new fortress, and if the Sultan had not wanted to keep the peace at all cost after a war in which the Ottoman Empire suffered so much,[7] and if he had wanted to give consent to the solicitations of the Crimean Tartars,[8] he perhaps would have already again taken up war with the Muscovites, for whom the goal would appear to be the acquisition of still more ports on the Baltic Sea; but they will find difficulty in this. The king of Sweden up until now had seemed far removed from a peace with the king of Poland, whom he would have wished to force out of his kingdom, having effectively on his side a large faction of Poles and Lithuanians who were so inclined. But the affairs seemed to have taken another turn, and it appears that the king of Sweden will be able in the end to resolve to deal with the king and the republic, wherein he would like to see the society opposed to the Muscovites. To this, however, I do not believe that the king of Poland will lend assistance, for the czar faithfully helped him and, whereas a perpetual peace was made between the Poles and the Muscovites a few years ago, to my knowledge the Poles have no reason at all to break it.

But let us leave the affairs of the world which scarcely interest China, and come back to the sciences. The analysis of the infinitesimals has given us the means, for myself and those who have advanced my methods, to resolve difficulties in which geometry has not previously succeeded. For example, no one had determined the figure of the catenary that Galileo had believed to be a parabola. Now, I have shown that by means of a very thin chain, like those used in clocks, suspended by its two ends, one will have logarithms ready made without calculation, as exact as the size, fineness, and flexibility of the chain permit. [9] I also do not know if I have spoken to you about my science of dynamics, or forces, where I have found the perfect way of determining them, deduced *a priori* from the sublime principles of real metaphysics, [10] and furthermore perfectly conforming to the experiments wherein I have not only found that it is false that the quantity of motion is conserved as the Cartesians believed, but that absolute force (understood as it should be) is [what is] conserved; whereas M. Mariotte, M. de la Hire and others of late have only [proposed] a conservation [of motion] of one thing with respect to another. I also found that there is as much action in one moment as in any other, either in the universe or in bodies which are conceived of as having contact only amongst themselves. But the quantity of motive action is quite different from what the Cartesians call the quantity of motion.[11] I demonstrate, moreover, that force is of the essence of corporeal substance and that it is the entelechy of the ancients, although it needs to be determined by the coincidence of bodies or by

the first disposition of God. It is therefore by the consideration of force that I vindicate the ancients against the too material philosophy of the moderns, to whom I concede that everything works mechanically in bodies, but I show them that they do not sufficiently consider that the very principles of mechanics come from something higher than that which is merely material or subject to the imagination.

But let us come to one of the principal subjects of your letter, that is, the relation of my binary arithmetic to the model of Fu Xi, who is believed to be one of the most ancient kings and philosophers known to the world and who is seen as the founder of the empire of the Chinese and of their sciences. This model being therefore one of the most ancient monuments of science found today in the universe, one that is more than 4000 years old, it would seem, and one that perhaps has not been understood at all for several thousand years, it is a very surprising thing that it perfectly matches my new manner of arithmetic and that I should have written you about my arithmetic at just the appropriate time, that is, just when you were applying yourself to the decipherment of these lines.[12] I assure you that had I not formulated my dyadics, I myself perhaps would have looked a long time at this system of the *gua* or lines of Fu Xi without penetrating their significance. For much more than 20 years I have had in mind this arithmetic by 0 and 1, in which I foresee marvelous means for advancing the science of numbers to a perfection which surpasses anything we have [yet attained] for it; but I held back from coming forward with it until I was prepared to show at the same time its great advantages. However, being as a thousand other occupations and meditations have hindered me from working on it, I finally disclosed it (although not yet in any printed book) in order that an idea of such consequence would not become lost. And I am delighted that [I have done so] precisely at a time when you were at a point to be able to put it to such good use as deciphering this ancient monument of China. In truth it appears there is some direction of providence in this. And I have no doubt at all that when this arithmetic is carried further, as it would soon be if there were some people both capable and inclined to work towards this, it will be a new means of reawakening the attention of the emperor and the leading people of China, although your decipherment of the *gua* of Fu Xi should already be sufficient in itself to astonish them.

I find that the square is the same thing as Fu Xi's circle and even serves to explain it, because the order is a bit confused in the circle. I begin in the circle with the bottom of the right side, that is with and etc. or with 0 and 1, or rather with 000000 and 000001. For it is necessary to look at the model as if the eye were located at the center, and to begin with the lines near the center. This is also shown by the characters, for they appear such as they are if one looks at them from the center, or rather if one turns the circle so that the part of the character which is the farthest from the center is the topmost; this demonstrates, then, the agreement with the characters of the square, which thus appear to be the same, such that with regard to the circle there is neither high nor low anymore than there is with respect to the sphere of the earth, where the farthest from the center is taken for the highest; and it is even possible that Fu Xi took this into consideration. But

you nevertheless were correct, my Very Reverend Father to write ἀνω [ano, i.e., above, up, top] and κάτω [kato, i.e., below, down, bottom] with respect to the square inscribed within this circle.[13]

But to return to the gua or linear figures of the circle, starting from 0 and 1 at the bottom of the right side we proceed up to the top of the right side, that is, up to (the line closest to the center always being considered as the first), that is 011111 or else 31. After that (instead of continuing in the circle and passing to the top of the left side and then descending) we start again from the left side on the bottom in order to go up from there; and now, on the left side on the bottom there is , that is 100000 or 32, apparently because in this way 0 and 32+0, or else 1 and 32+1, or 2 and 32+2 (which expressed binarily only differ in that the linear figure of one begins with ☐, or 0, and that of the other with ☐, or 1) are face to face one to the other while going up from the bottom to the top on both sides. Perhaps, also, it was wished that this would serve to provide a bit of a puzzle, and to show some apparent difference between the circle and the square. Finally, moving up from bottom to top on the left side, we finish with , that is 111111, or 63. With regard to the inscribed square, the order there is perfectly natural, beginning with

000000 000001 000010 000011 etc.,
0 1 2 3

where it should be noticed that the order of the numbers proceeds from left to right in each line or row of numbers, and the lines or rows follow from top to bottom; and in each case this agrees with the conventions that we follow in writing in Europe, as opposed to the custom of the oriental [peoples] and the Chinese. It is true however, that in each number it is necessary to count the lines from bottom to top, perhaps in imitation of the correspondence to the circle where we start from bottom to top, beginning with that which is nearer the center. The correspondence of so ancient a monument to European writing reminds me of having read that in some very ancient oriental monuments, it seems that one also read in the style of Europe. This would be, if I am not mistaken, in the unusual writing which is found in the ruins of Persepolis or of Chehil Minar[14] where the writing consists of lines and triangles and could well predate the monarchy of the Persians.[15] This ancient monument is spoken of in the accounts of Figueroa, ambassador of Spain in Persia,[16] and in those which are found among the relations of voyages [edited] by the late M. Thévenot,[17] and elsewhere.

But let us return to the dyadic calculus which makes use only of nothingness and unity. You have keenly sensed one of its principal uses for religion, which is that it is a potent symbol of the Creation,[18] that is, of the origin of all things from God alone and from nothingness, without any preexistent matter at all;[19] and that the non-absolute but respective nothingness—that is to say its limitation—is found essentially in creatures in proportion to their imperfection, this limitation being nothing other than a negation of the further development of pure reality or of pure action,

as when a circle is bound by its circumference which gives to it its *non-plus-ultra* [i.e., its outer limit, ultimate definition]. And I believe that the learned men of China, when they become well versed in this idea and above all when they see that Fu Xi's ingenious design conforms to that of our own [tradition], will be sufficiently inclined to believe that this great man had furthermore wished to represent God, maker of things, and the creation by which He drew those things out of nothingness. Thus, this could be one of the most important points of your catechism, taken from the classical authors of China and worthy of being explained to the emperor himself.[20]

As to the usage of the dyadic expression of numbers, that is by 1 and by 0, for the perfection of numerical science, I have demonstrations that can persuade us that it is a way to bring this science well beyond its former state; for I have found that squares, cubes, and other powers of natural numbers also have [repeating] periods in their columns, as do natural numbers, which provides a wonderful facility for determining them and for setting up tables for them with ease, and for uncovering relationships unknown until now.[21] I also find that this will be of great utility in expressing incommensurable geometric dimensions by series of whole numbers which approach infinity, whereas what is lacking in Ludolph van Ceulen's[22] calculations for the circle (for example), is that there is no formula by which one might continue further. But having no leisure at all for furthering these researches, in the event that I do not find anyone who has the capacity or inclination to assist me, I will leave to posterity the benefits that can be reaped from them, benefits greater in extent than can be imagined by those who consider only the superficies of things.

With regards to the usage of the double geometric progression in the arrangement of ideas, of which Your Reverence also has made reference, experience has made me realize the great use of dichotomies in the formation of ideas.[23] It is true that one can make dichotomies in different ways, but they lead to the same end, that is, to the same species by way of different subordinate classes. And it is this that makes it possible for the same terms to receive different definitions which, however, analysis will show to be congruent in the end. I believe that there are few people who have worked more on the arrangement of ideas and who have a greater scheme for them, but in the midst of occupations and distractions, I cannot imagine how to clear up my disarray without being assisted by people who have more vigor and leisure than I. However, if God grants me a few more years of life, I will make an effort to put some order to this work which appears to me one of the most important that one might undertake, since it would provide for reasoning a *filum Ariadnes* [i.e., thread of Ariadne], that is, a tangible method in the form of a calculus to guide us, whether in matters of judgment or invention.[24]

I suspect that Fu Xi assigned the 64 numbers (either uncompounded or expanded to 128 or even further) to the terms he conceived as the most basic, and that he provided for each of these terms its own character which also designated its number or rank; and that then from these most simple

and principal terms and characters he formed the others by adding small strokes. But in the course of time these characters were altered, as much by the nature of popular usage which little by little changes the strokes (as can be seen in comparing the ancient writing of some languages with the modern), as by those who, no longer knowing the logic or the method of the characters, adapted them according to their own whims, which were often based on metaphors or some other more tenuous connections. If we well understood Chinese literary history—which in writing you I formerly hoped to see established according to good critical methodology in order to differentiate between the ancient and the modern—then we would be better able to pass judgment. It is also likely that different princes or philosophers made different reforms of the characters for the purpose of making them better; but as they did not always follow the same laws of grammar or etymology, as it were, their origins were finally entirely obscured, even more than in the case of the broken lines of Fu Xi wherein the meaning had been lost but has just been recovered by us.[25] I would be very glad to learn the signification of the Chinese characters attached to each number in Fu Xi's model, and I beg Your Reverence to send me the explanation if it will not cause you too much trouble. Apparently they are modern explanations or comments which were fabricated without knowledge of the true usage of the model, that is, the calculus by means of 0 and 1.[26]

If you find that the Chinese of today have no knowledge at all of this calculus, Your Reverence might then, in the presence of the emperor himself and the principal scholars of China, boldly attribute the success of deciphering the true meaning of Fu Xi's model, and its broken lines, to the aid of a new discovery coming from Europe in the manner of calculating by 0 and 1. This, in my opinion, will not be perceived to be of little consequence, and should raise for the Chinese the esteem of European sciences and, by consequence, our religion. This also will place in them great expectations for the mysteries still hidden therein that remain to be uncovered, and will even give us an open field for devising a new characteristic which will be seen as a extension of Fu Xi's [model] and which will be the starting-point for the analysis of ideas and for this marvelous calculus of reason for which I have a preliminary plan. This secret and sacred characteristic would also give us the means to introduce to the Chinese the most important truths of philosophy and natural theology in order to facilitate the way to the revealed [theology]. And as new or different as it may be from their own, it will be acceptable thanks to Fu Xi and will eventually become, as it were, a language exclusive to the highest class of scholars, those most enlightened and most closely tied to Fu Xi, meriting even separate [research] societies and colleges. Thus, this discovery could have great ramifications for the whole of the Chinese empire if one knew how to take advantage of it either where you now are or moreso in Europe. [crossed out part, ck. orig. ms. p. 33a]

For now that we have found the key to these lines and the Chinese cannot at all doubt that we have succeeded in this, we will be incomparably more accredited in their eyes. The Characteristic that we can build in this manner will reduce ideas into numbers, and will at the

same time have the advantage of subordinating them to calculation just the same as numbers, something that is beyond all that we have promised ourselves in these matters, and that many people, not without some semblance of reason, will believe to be impossible. Moreover some means of presentation will be necessary, for it is something not easily imagined, and it should carry with it its own proof. You will judge, my Very Reverend Father, whether this should not inspire those whose interests lie in the perfection of the workings of the human spirit, and above all in the progress of virtue and the true religion. But the present war in Europe, and perhaps the importance and the very magnitude of this discovery of the general characteristic, greatly diminishes my hope of seeing it carried out with the assistance which would be necessary, for one wonders whether providence does not wish to reserve it for another time when men will be more suitable to attract the graces of Heaven.

You will see by this, my Very Reverend Father, that it would not be enough to [simply] set up several divisions and subdivisions in order to arrive at the species [of things], for this would not at all furnish an analysis capable of supplying the properties of those species by means of the characteristics that would be attributed to them by virtue of these divisions, in which it is further necessary to make a great distinction between "confused" ideas and distinct ideas.[27] For the characteristics of distinct notions could be exact right from the start, but those of confused [notions] can be nothing but provisional and proportionate to that which has been recognized as distinct. It is left for posterity to perfect them according to new experiments.[28] And this observation is important, as much for not promising that which is beyond the present power of man and leaving us open to vanity, as for knowing exactly what there is to do. As to colors, experiments with prisms show that in addition to white and black, the principal colors are yellow, blue, [and] red—which with blue makes purple—and that green is but a combination of yellow and blue; and [these experiments also show] that this is determined by the bending of the rays during refraction, and not by the mixture of only black and white.[29] But this is said in passing, and you have only cited colors as an example of clarification where the applicable rule is exemplorum non requiritur veritas. 30 I also would not want to jump from the third degree to the sixth, for neither is the method very well founded, nor will it tolerate these sorts of jumps without injury. The eight degrees that the scholastics give to the qualities (for I do not remember that Aristotle used them) 31 came from their believing two of the four elements to always have a common quality, and wishing to place four degrees within each; for example four degrees of heat in fire (as per the chemists) and four others in air.

A certain Englishman named Suiseth, 32 called the Calculator par excellence who lived more than 300 years ago, refined to an extreme degree the concepts of intensity and remission, that is, the degrees within the qualities. But this manner of calculation does not at all seem to suit our characters. Things as well as ideas do not differ *gradu tantum* [i.e., only by degree], but in many other ways.

It seems, however, that the eight *gua*, or eight linear figures which are considered fundamental by the Chinese, could lend belief that Fu Xi himself had Creation in mind in making everything come from the One and the Nothing, and also that he even extended the correlation to the story of Genesis. For the 0 could signify the void which precedes the creation of heaven and earth, then follow the seven days, each of which marking that which existed and was found made when the day started. At the beginning of the first day there existed 1, that is God. At the start of the second, 2, heaven and earth being created during the first. 33 Finally, at the beginning of the seventh there already existed everything; this is why the last is the most perfect and the Sabbath, for on that day everything is found made and full; thus 7 is written by 111 without 0. And it is only in this manner of writing by 0 and by 1 that the perfection of the septenary which is considered sacred may be seen, where it is even more remarkable that its character has some correlation to the Trinity.[34]

I believe I have written to you before that a theologian from Berlin who died a few years ago named Andreas Müller, of Pomeranian nationality, rather deeply learned in oriental literature and in the knowledge of languages in general and who provided some excellent notes to the *Cathay* of Marco Polo,[35] believed he had found some key to Chinese characters, according to what he published in a small printed paper; but it is known that he demanded a considerable sum of money for disclosing it. I introduced him to the Very Reverend Father Grimaldi[36] who wished to speak with him while going from Vienna to Poland, but Monsieur Müller did not go to the meeting. He was a man of much merit but he was equally capricious. The late Imperial Elector of Brandenburg[37] seemed disposed to give him whatever he wanted, but he was not at all tractable and even withdrew from Berlin to his homeland. He threatened to burn his papers before dying and it is said that he kept to his word. It does not at all seem that he knew of the calculus hidden in the *gua* of Fu Xi.; nor was he enough of a mathematician to perceive such an arithmetic. However, I would not know how to easily persuade myself that a man of such merit would boast of a secret unless he thought it had at least some likelihood. Thus I suspect that he glimpsed some relationship between the ordinary characters for numbers of the Chinese and their root characters for things, and perhaps also [had] some observations concerning the small additions [i.e., strokes] which modify the root characters in order to make derivative ones, or perhaps even composite ones. But this is something Your Reverence and the Reverend Father Visdelou will judge better than I could. For it would be necessary for me to have good dictionaries of the Chinese characters with a European explanation in order to ponder it.

Your Reverence informs me that such dictionaries cannot be found where you are. But it must be that someone hides them from you because surely they exist, and even in print as I recall. My evidence is based on the catalogue of oriental manuscripts of the incomparable Golius,[38] which were sold by auction in Holland to my great astonishment and regret about six years ago. For they were a public treasure worthy of the greatest king, and they should have been conserved by the University of Leyden with the aid of the heads of State or the Society of the East Indies.

Now, in this catalogue I remember that there were two Chinese dictionaries, one Chinese-Spanish,[40] printed I believe in the Philippines, and I do not remember whether the other was Portuguese or Flemish. I hope I have kept this catalogue, and when I return to Hanover I will try to find it in order to clearly indicate everything to you. One of the two surely explained the characters, but I could not say whether they both did, or whether one perhaps concerned itself only with the Chinese language. I have been told that most of these pieces went to England.

I do not know what to say about the hieroglyphs of the Egyptians, and I find it hard to believe that they have any relationship to those of the Chinese. For it seems to me that the Egyptian characters are more commonplace and bear too much resemblance to perceptible things, such as animals and other things, and by consequence to allegories, whereas the Chinese characters are perhaps more philosophical and appear to be based on more intellectual considerations, as might come from number, order, and relations; thus, they are nothing but disassociated strokes that do not point to any resemblance with any particular sort of thing. I know that some have believed that the Chinese were a colony of the Egyptians, based on the supposed agreement of the characters, but this has no likelihood at all.[41] There is an ancient book on the hieroglyphs of the Egyptians by one Horapollo.[42] One can also see in Ammian Marcelin,[43] if I am not mistaken, the explanation of the characters of an obelisk which is still found in Rome. Pierius Valerianus,[44] Father Kircher in his *Oedipus Aegyptiacus*, *Obeliscus pamphilius* and other works,[45] and Laur. Pignorius[46] in *Mensa Isiaca [...] explicata* (which is an ancient work full of Egyptian hieroglyphs that I have learned is at present in the library of the Duke of Savoy), tried to shed light on these hieroglyphs. I would advise you to obtain and use these books if I believed that they could serve in any way in understanding the characters of China. The late Monsieur Thevenot,[47] librarian of the king, had a book published on the characters once used by the Mexicans. But I believe they do not have any relation to either those of the Chinese or those of the Egyptians.

With regard to the Very Reverend Father Grimaldi, of whom I have just spoken on the subject of the Chinese key of the late Monsieur André Müller, I am delighted that he remembers me favorably, but am upset that his health is not more vigorous. I beg Your Reverence to indicate to him how much I honor him and how much I am concerned about his health. I am further obliged to send greetings to him on the part of the Very Reverend Father Vota,[48] confessor to the king of Poland, who has just left Berlin where he had been for a short while as he also had been at Hanover where he had accompanied the Queen of Prussia. He now has left to return to the king, his patron, in order to be with him during Holy Week. He told me that the account of the voyage which Father Grimaldi had sent him[49] was lost last year along with other papers and baggage when the army of the king of Poland was set in an uproar by that of the king of Sweden, both kings being present, when the Poles and then the Saxons too were seized by panic and terror, for there is nothing so contagious as fear. He would strongly wish to obtain this account as also would I. Thus, if the Very Reverend Father Grimaldi will grant us this favor, I ask Your

Reverence to send it to me, as well as the astronomical observations that I hope can be obtained from him and from other friends.

That which Your Reverence communicates to me in the last two pages of your letter, concerning the signs of the cult of a Sovereign God and the traces of the true revealed religion in ancient China which are found in their characters and in their classical books, seems important to me. I have always been inclined to believe that the ancient Chinese, like the ancient Arabs (witness the book of Job) and perhaps the ancient Celts (that is [ancient] Germans and Gauls), were far from idolatrous and moreover were worshippers of the sovereign principle. Cluverius, no doubt an excellent man, in his *Germania Antiqua*[50] went so far as to believe that the old Germans had had some knowledge of the Trinity. I am not at all of his opinion. However, I have not seen anyone find fault with him for this, and I find it strange that there has been so much commotion made against you fathers who have written that the ancient Chinese had the true religion. What harm is there in this? If it were false, is this an error which would give birth to dangerous consequences?[51] Not at all. [52] [NOTE CROSSED-OUT PORTION: “Would to God that it were true, and that the Chinese believed it.... Putting emotions aside, that [criticism] could only come from the prejudices of some foolish men who, full of themselves and of the special favor... that they conveniently believe that God has accorded them (quite similar to the young lady from the land of Lunebourg [53] who was otherwise sensible but who looked on the world contemptuously because she believed herself the wife of Jesus Christ who had appeared to her), and lacking... sufficient charity, that is to say good will, are the antipodes of Caelius Secundus Curio[54] who wrote *De Amplitudine... Regni Coelestis*, and... are quick to damn even infants who die without having been baptised. { Which shows that they do not think enough about the wisdom and kindness of God. } Which shows the strange idea that they must have about God { when they do not give it sufficient thought}..., were they to sufficiently consider the consequences of what they attribute to Him.] Would to God that the Chinese did not have anything else.

For the remainder, I beg you, my Very Reverend Father, to remember the Lord's Prayer in different languages. If one could add to it a little catalogue of other most commonly used words, this would be so much the better. Likewise, for you to remember the geography [i.e., geographical account] of Tartary with comments on Monsieur Witsen's map which you certainly will have along with you. And likewise, to remind Father Grimaldi and others to get for me some old Chinese astronomical observations; and also several good modern ones, especially if you have any of the eclipse of the year..., which would serve to firmly establish the meridian of Peking.[55] I would also like to know whether something more definitive could not be learned about the ancient conjunction of all the planets, according to which an ancient emperor began the cycle which the Chinese use in their chronology.[56] I repeat also my entreaties on the manufacture of paper in China, and on the method of enclosing air in leather or in some other material so that even when pressed quite hard, it does not at all leak out through the pores. And likewise regarding the Hebrews who some say have been in China from time immemorial, about which I

believe to have written to you,[57] in order to learn if it is possible that they have the complete Old Hebrew Testament or only the Pentateuch and Joshua as do the Samaritans;[58] and what is their script, whether it is Samaritan which is the most ancient, or that which the Jews still use today which is believed to have been brought from Babylon. Likewise, whether they have [vowel-] points and accents or not.[59] And if I asked for anything else, either from Your Reverence or from Father Grimaldi, I repeat it and beg you to consider it at your leisure with your friends, and likewise regarding the questions of Monsieur Schrokius or the Late Father Kochanski. Nor should you slight the question that I believe I posed concerning maladies, among others whether gout and stones are as frequent in China as in Europe, and likewise smallpox and the other variola. For all these reports are of consequence. Likewise, whether the Chinese have remedies as effective against some ailments as quinine is against fever[60] and ipecacuanha is against dysentery. I also remember having read in some books which speak of China, and if I am not mistaken in the atlas of Father Martinius,[61] that the Chinese have a method of producing silk twice a year, in the same place and from the same mulberry trees. It would be good to know if it is true and in what it consists.

Finally, I pray God that He grants you sustained health, and that He gives you success always in your fine designs. Being with zeal,

My Very Reverend Father,

Your

P.S.

I ask pardon of Your Reverence for being aided by another hand as I am so occupied and am not too well. I looked at (after writing and having the above copied) Fu Xi's characters in the *Confucius* of Paris that was printed from the accounts of the Reverend Father Couplet, where I find only the [groups of] four and eight *gua* well arranged; the 64 are in a confused order.[62] But conjectural significations were apparently added, such as air, water, etc., that the linear figures are believed or were believed to signify. Is it perhaps that the characters around the imprinted circle which you sent me also signify that which is noted in the *Confucius* of Paris? I ask you to examine it, my Very Reverend Father, and to let me know. For I do not at all doubt that you have in Peking this book that was printed in Paris.

I must also add, Your Reverence, on the subject of Chinese dictionaries, that since writing this letter I believe I have found here in the library of the king of Prussia one of the two dictionaries mentioned in the catalogue of Golius' manuscripts, the one which is in Chinese and in Spanish.[63] I see that this dictionary, which is there in manuscript, strictly speaking explains the Chinese language but not the characters, even though to each word of the language is added its Chinese

character. Some one spoke to me formerly of a great dictionary of the characters called *Hai pian* [i.e., 海篇], or ocean, which no doubt is readily found in China but is not explained in any European language.[64] However, that other dictionary, or vocabulary rather, with illustrations of things, of which I spoke about in my last letter[65] and which Your Reverence gives me hope of obtaining, appears to me to be singular and serviceable.

If Your Reverences give me the grace of sending me something, either of physics or mathematics, the arts, languages, or geography, I ask Your Reverences either to address it to me or indicate very clearly to your friends in Paris that they should communicate it to me, otherwise I will have difficulty in availing myself of it.[66] And I have not as yet been able to learn anything in detail, nor receive word of all that the Very Reverend Father Fontaney brought, and about which I again inquire;[67] perhaps these things have already passed into other hands.

As this father wrote me a word from Canton after his return,[68] and in a very obliging manner responded to the letter which I gave myself the honor of writing him when he was in Paris, I here answer by the subjoined which I would ask Your Reverence to deliver to him.

The title of the Chinese-Spanish dictionary of which I have just spoken is *Vocabulario de letra china con la explicacion castellana hecho con grand propriedad y abundancia de palabras por el Padre Don Francisco Diaz de la Orden de Predicatores ministro incansable en esto Reyno de China*.

Apropos of mules which reproduce in Tartary, someone suggested a passage to me from Strabo which says as much of the mules of Cappadocia.[69]

The [inscribed] monument of the Christians who were in China more than one thousand years ago, [of which the inscription was] published by Father Kircher—is it still extant?[70] I would like a report which is well supported by Chinese evidence itself, taken from their chronicles of these times, in order to shut the mouths of those persons who dare to say that the modern Christians forged it; which does not seem believable to me when I consider its contents.

Previously, My Very Reverend Father, you sent me a Lord's Prayer in the language of the Chinese Tartars, but it seems to me that you said you had composed it.[71] Thus, if the one which is used among the Christian Tartars is different, I ask you to send it to me among the others.

Since I wrote this, I found in the library of the king of Prussia another small Latin dictionary with Chinese characters and words, printed in China itself apparently through the attentions of the Reverend Jesuit Fathers; but this small dictionary only contains the most necessary words.[72]

I ask you, finally, to forward the attached to the Very Reverend Father Fontaney. It answers the letter he gave me the honor of writing from Canton on the 15 September 1701, which I received several months before yours. I am sending him a catalogue of questions, and in asking him to consider it at his convenience and to give me some insight thereupon, I ask you to collaborate with him as if it were also for you, which it is in effect.

[APPENDIX]

Excerpt of my letter to the Reverend Father Fontaney:

My Very Reverend Father,

I received the honor of your letter from Canton of the 15 September 1701 and I thank you very much for your remembrance of me, which I hope to benefit from again as soon as you have the occasion and the leisure to send me some insights. I ask Your Reverence, along with the Very Reverend Father Bouvet and other friends, to think about the attached questions, in addition to those [questions] that I have sent to that Father both now and previously. I congratulated him for deciphering the enigma of the lines of Fu Xi with the aid of my binary arithmetic. This discovery will prove to be of consequence to them there will not last long after they have drawn from us our sciences. Thus I hope that we will work while we have the means to have Europe also profit by the sciences of China, more than we have until now. And [I hope that] we will try to interest the emperor in having descriptions made of all the arts, as much for providing the means to perfect them as for assuring that the Chinese keep nothing hidden from the Tartars. This will also be the way to perfect the dictionary which the emperor is having made, for the terms for the arts could be entered therein.[73] Many persons have not been pleased with me, as much among the Protestants as in France, for my taking the side of your missions. The Pope acted wisely in suspending his judgement on the subject of the religion of the ancient and modern Chinese,[74] as one must not condemn persons not understood. However, whatever may be the opinions of some of them, it is significant that the emperor has declared himself in favor of the truth.[75] Someone told me that since the defeat of the king of Eluth,[76] it is now possible to go to China both from Persia, by way of the Usbeks, and from the country of the Mogul. But if one could be sure to go as easily as did you by sea, then that path would be preferable to all others.

Endnotes

1. Leibniz is responding to Letter I, having not yet received Bouvet's 1702 letter (K).

2. Leibniz here further details reasons for his ongoing fear of a one-sided intellectual exchange with the Chinese. See Letter B, n. 31.

3. Baron Francis Mercurius van Helmont (1618-1698), son of the famed medical doctor and alchemist, Jan Baptista van Helmont (1579-1644), became a close acquaintance of Leibniz during the last years of his life when he was a guest of the Court in Hannover. Leibniz even ghostwrote a book for him in Latin on the first four chapters of Genesis. A trained alchemist and Kabbalist, he was knowledgeable in many areas, including painting. Leibniz sought his advice in many areas – practical and theoretical. For further details on this fascinating figure, see A. P. Coudert, *The Impact of the Kabbalah in the Seventeenth Century: The Life and Thought of Francis Mercury van Helmont (1618-1699)* (Leiden: Brill, 1999). For details on van Helmont's relationship to Leibniz, see *ibid.*, "Leibniz and the Kabbalah," pp. 309-329. For a detailed study of Leibniz on the Kabbalah, see A. P. Coudert, *Leibniz and the Kabbalah* (Dordrecht: Kluwer, 1995).

4. The War of the Spanish Succession (1701-1714) was occasioned by the death of the Spanish King, Charles II, who was childless. A Grand Alliance was formed, led by England and Austria, to challenge the French claim to the Spanish throne. Prussia and the other German states sided with the Grand Alliance against France.

5. On the Great Northern War, see Letter H, nn.. 14; 17.

6. Nöteburg (present-day Shlissel'burg) is on the Neva at Lake Ladoga. Until the founding of St. Petersburg in 1703 (as a symbol of Peter the Great's early victories over Sweden) to its west, Nöteburg was the major city in Ingermanland/or Ingria, the area south and east of the Gulf of Finland.

7. The forces of the Ottoman Empire were defeated at the gates of Vienna in 1683 by the Austrians (with the help of the Poles), at the beginning of "The Great Turkish War" (1683-1699). Further defeats of the Turks during the war resulted in the loss of most of Hungary to the Austrians. Sultan Suleiman II died in 1691, so Leibniz is referring to recent past events (as he often did in sending news about Europe to Bouvet), rather than present ones. In early 1703, at the writing of this letter, Mustafa II had been the ruling Sultan since 1695, something that Leibniz would obviously have known.

8. The Tartars, who call themselves "Tatars," are Turkic-speaking Muslims. Over several centuries, the Tartars on the Crimean peninsula and southern Russia were gradually brought under the rule of the Czar.

9. Leibniz had already discussed this solution in Letter H, n.10.

10. Leibniz briefly mentioned his disagreement with the Cartesians on the need and nature of force to explain matter and motion for the new science of mechanics in Letter B, nn. 33-35. The

“sublime principles of real metaphysics” refer to the need to reintroduce the Aristotelian notion of substantial forms into mechanics as the concept of force. In Leibniz’s words, “... having attempted to examine the very principles of mechanics in order to explain the laws of nature we learn from experience, I perceived that considering *extended mass* alone was not sufficient, and that it was necessary, in addition, to make use of the notion of *force*, which is very intelligible, despite the fact that it belongs in the domain of metaphysics.” AG 139; G IV 478. Leibniz also discussed his theory of dynamics at some length in Letter H, nn. 31–33.

11. Leibniz repeats his remarks from Letter H, nn. 28–29, but adds the name of the physicist and member of the French Academy of Sciences, Edmé Mariotte (1620?–1684). Mariotte, Prior of Saint-Martin-sous-Beaune, near Dijon, had written a work on Cartesian mechanics, *Traité de la Percussion, ou choc des corps, dans lequel des principales règles de mouvement sont expliquées et démontrées par leur véritable causes...* (3rd ed.: Paris, E. Michallet, 1679).

12. Leibniz was indeed so “surprised” by “the relation of my binary arithmetic with the model of Fu Xi,” that even before he responded in this letter to Bouvet’s discovery, he had already discussed it at length in several communications. Leibniz devoted a whole letter to it to Father Vota, (on Vota, see p. 29 and n. 48 below in this letter), apologizing at the end about going on and on about a matter “which is perhaps not to your taste.” (Z 291) A few days later (7 April 1703), he added a section on this “Chinese connection” to his inaugural presentation on binary arithmetic to the French Academy, emphasizing this addition in an accompanying letter to its director, Abbé Jean Bignon, and suggesting it replace his earlier paper on it. (Z 292-301) Understandably, Leibniz discusses the importance of binary arithmetic for justifying the doctrine of Creation to the pagan Chinese in his letter to Father Vota, but omits it in his paper to the French Academy of Sciences.

13. N.b. See Alan's yellow note page on this.

14. Persepolis was an ancient capital of Persia, now in southern Iran, not far from Shiraz. Chehil Minar (lit. “forty minarets”) was a Persian name for Persepolis in Leibniz’s time. Later in his *Theodicy* (1710) in his long discussion of Zoroaster and the religion of the ancient Persians (§137), Leibniz specifically mentions the sculptured representations of their ceremonies in “the splendid ruins of Persepolis or of Tchelminaar (which means forty columns).”

15. Leibniz is referring here to the cuneiform script of Old Persian. At Persepolis, it usually appeared in trilingual form, along with Elamite and Akkadian (also written in cuneiform). The decipherment of the script was a collective effort, stretching over many years (1798-1932). Twentieth-century scholarly opinion agrees with Leibniz that this system of writing may indeed antedate the monarchy of Darius I and Xerxes. See R. G. Kent, *Old Persian: Grammar, Texts, Lexicon* (2nd ed. rev.; New Haven: American Oriental Society, 1953), pp. 9-12.

16. Don Garcia de Silva y Figueroa (1574-1628) visited the site in question in 1618-1619, during his diplomatic stint in Persia. His journal was translated as *L'Ambassade de don Garsias de Silva et Figueroa en Perse*, by F. de Wicquefort (Paris, 1667). (Our thanks to David Stronach of the University of California at Berkeley for the information and references in this and the previous notes on Persia.). Leibniz also mentions that an ambassador of Holland had had the ruins sketched by a painter at great cost; see *Theodicy* §137.

17. Melchisedek Thévenot (1620-1692), European traveler and scholar of languages, corresponded with Leibniz until his death. From 1684, he was the Royal Librarian in Paris. He compiled and edited the accounts of many other travelers in a two-volume work, *Relations de divers voyages curieux qui n'ont point été publiées, et qu'on a traduit ou tiré des originaux des voyageurs français, espagnols, allemands, portugais, hollandais, persans, arabes...* Paris: Thomas Moette, 1696. The materials on Persia are entitled, “Description des antiquités de Persépolis” (trans. from the English).

18. Bouvet further elaborated upon the binary analogy to creation that Leibniz proposed in Letter I, applying his own theory of the binary origins of Chinese characters to demonstrate the possibility of a universal characteristic capable of expressing the nature of Creation as well as all created things. See Letter I, *passim*. On Leibniz’s views on a universal characteristic, see Letter H, n. 40.

19. Leibniz repeats his (Neo-)Platonic theory of creation, which equates the limitation of all created things with the imperfection or negation that he briefly outlined in Letter H. See nn. 19; 22.

20. Leibniz in effect articulates Bouvet’s Chinese strategy: if Christendom can demonstrate to the Chinese that their most ancient characters have an original meaning that has been lost to them, but recovered by the Christian West, then the path to their conversion would be considerably eased, since in converting to Christianity, they would be *recovering* their own roots, rather than abandoning them for a foreign religion. Leibniz expressly compares this strategy for converting the Chinese to doing the same with the Jews: i.e., “to be able to interpret their ancient books better than the [Chinese] erudites themselves. Who does not know in our own day that Christian scholars are much better interpreters of the most ancient books of the Hebrews than the Jews themselves.?” See his “On the Civil Cult of Confucius,” in CR 64; WB 252-254. Leibniz used this strategy himself as a Protestant against the Catholics, demonstrating that the latter are not true to the texts of the Church Fathers; see Baruzi, p. 373.

21. See the *Demonstratio quod columnae serierum . . . sint periodicae* of November 1701, GM VII: 235-238. For various details and references on Leibniz's usages of binary notation, see Z 178-181 *et passim*.

22. Ludolph van Ceulen (1540-1610), Professor of Mathematics and Military Science at the University of Leiden, had computed the value of π (known also as Ludolph's or the Ludolphine number) to 35 decimal places. Even though π has now been calculated to over two billion places with the aid of super-computers, no algebraic formula has been found for determining these calculations. On Ceulen, see *Dictionary of Scientific Biography*, ed. C.C. Gillispie (New York: Scribner's, 1981), III 181a-b.

23. Leibniz is referring here to Bouvet's efforts in Letter I to develop some sort of universal characteristic based on dichotomies or binary distinctions. (note 18 above). The discovery by Bouvet of the binary basis of the ancient Chinese characters of the Yi-Jing helped revive Leibniz's lifelong efforts at developing the proper characters for a universal characteristic or, as he often called it, an "alphabet of human thoughts." It might also be the basis for a *Key to Chinese* (Clavis Sinica) that the West had been looking for as a means of easily and systematically deciphering Chinese characters. Hopes for such a "Key" had been dashed after the death of Andreas Müller in 1694 (see Letter B, n. 8). If these ancient characters are the basis of the Chinese ideograms, this would means that the Chinese language had an arithmetical basis that could serve as the model for a universal characteristic or "calculus of thought." See Yuen-Ting Lai, "Leibniz's Studies of Chinese and Perennial Philosophy," *il cannocchiale* 10 (January-April 1999), p. 127.

24. Leibniz is referring to his earliest work, his *Dissertation on the Art of Combinations of 1666 A VI: N. 8*; selections in L 73-84). Leibniz ably described his goal in a struck passage in Letter B, n. 36. Leibniz reiterates here the main scientific and philosophical purposes of his ideal "calculus of thought" (besides serving possibly as an easily learned and communicable universal language): viz., as a method of judgment that could check the validity of arguments and as a method of invention that could uncover new truths and arguments.

25. As Leibniz put it a few years later "Fohi, the most ancient prince and philosopher of the Chinese, had understood the origin of things from unity and nothing, i.e., his mysterious figures reveal something of an analogy to Creation, containing the binary arithmetic (and yet hinting at greater things) that I rediscovered after so many thousands of years, where all numbers are written by only two notations, 0 and 1." "Remarks on Chinese Rites and Religion" (1708), CR 373; GP II 383. The "greater things" alluded to may be the universal characteristic that Leibniz had sought since his youth.

26. Leibniz is referring here to the traditional interpretations of the 64 hexagrams of the *Yi jing*, or *Book of Changes*, which was viewed both as a book of oracles and of wisdom. For background as well as interpretations of each hexagram, see the English translation of R. Wilhelm's *The I Ching or Book of Changes*, trans. C.F. Baynes (3rd ed.; Princeton: Princeton University Press, 1967). For a description of how one went about consulting the *Yi jing*, see CR 20-22.

27. Leibniz succinctly summarizes this traditional epistemological distinction in a passage (later stricken) from his *Discourse on Metaphysics*: "When I know only the possibility of a thing by experience, the idea/knowledge which I have of it is confused; because all that exists is possible. It is thus that we know bodies and their qualities. But when I can prove the possibility *a priori*, this knowledge is distinct." §XXIV (AA VI iv, p. 1567, note to l. 18; LG 41, note a)
Leibniz is here criticizing Bouvet's efforts at a characteristic outlined in his last letter because he relied on unanalyzed, empirical distinctions (e.g., colors), whereas the characters of a universal characteristic (UC) or species must be analytically transparent, so to speak. They must capture the specific properties of the object denoted in a way that allows us to analyze it, thus being able to reduce it to its basic elements as well as to detect its distinctive nature when it is found in combination with others. Thus Leibniz continues (this time in the actual text of the *Discourse*) with an example of distinct knowledge: ". . . such is the knowledge of an assayer, who discerns true gold from false by means of certain tests or marks which make up the definition of gold." *Ibid.*, p. 1568; LG 41. Leibniz's example from chemistry is apposite: chemical tables and formulas are an excellent example of the basis of a UC, since they allow us to detect the specific properties (alone or in compounds) of an element as well as discover its other possible compounds, because of the number and position of its electrons, etc.

28. These "new experiments" would enable us to reduce all things to their natural "elements," so that just as we can determine possible and non-possible compounds in chemistry, so could we do the same with the proper UC, using the deductive laws of logic and physics to detect false arguments and erroneous experiments. In the manner that mathematicians resolve an argument, scientists and philosophers could likewise "sit down with their pens and abaci . . . and say to one another 'calculemus!' (let us calculate!)." A VI iv, p. 193; also A I iv, p. 315.

29. As early as 1666, Isaac Newton (1642-1727) used a glass prism to diffuse a beam of sunlight into the colors of the spectrum. While Newton's momentous discovery on the nature of color was incorporated into his magisterial work, *Opticks*, published in 1704, he had publicly communicated his findings much earlier in lectures and letters, usually to the Royal Society of London. The latter, addressed to the Secretary of the Society, Henry Oldenburg, were published in the *Philosophical Transactions* of the Society at the time. In Letter N, Leibniz says that Newton's work "entirely upsets the the ideas which Descartes gave us on the nature of rays of light." On Descartes theory, see Letter N., n. 17.

Before Newton, black and white were treated as two qualitatively different poles of the spectrum, with each “color” in between being made up solely of a certain gradation or mixture of the true colors, black and white. Newton’s experiments demonstrated that black was not a distinct color, but the absence of all color; white, previously thought to be the purest or lightest of colors, was nothing but a mixture of all other colors! Thus black and white are not true colors at all, while those “in between” are in their mathematically measurable gradations in effect the true primary colors. By denying the polar opposites of black and white – like those of light and dark, hot and cold, etc., i.e., any qualitative distinction -- Newton effectively destroyed the scholastic Aristotelian tradition of degrees and affinities of basic qualities as a form of scientific explanation. For an excellent discussion of Newton’s contributions as well as copious references to the appropriate written sources at the time, see P.S. Westfall, “The Development of Newton’s Theory of Color,” *Isis*, LIII (1962): 339-358.

30. "Truth is in no need of examples". H. Zacher refers to a fragment from April 1703 in L. Couturat (C 327) in which Leibniz uses this phrase in reference to Bouvet's example. (See Z 284)

31. According to Aristotelian theory – as understood by the Scholastics – physical changes come about because each of the four basic elements, earth, air, fire and water, can be qualified or changed to some degree by a combination of two of the four basic qualities, hot, cold, wet and dry.

32. Richard Swineshead or Suiseth, known as the Calculator, flourished in the mid-14th century. He is best known for his attempts to apply mathematical and deductive techniques to calculating physical changes. In the first chapter of his most famous treatise, *Calculator* . . . (Venice 1520), employing the traditional Aristotelian explanations of qualitative changes (mentioned in the previous footnote), he discussed the concept of intensity (or lack thereof: i. e., remission) of a specific quality. See Chapter XXIII, “Calculator and the Rise of Mathematical Physics,” in L. Thorndike, *A History of Magic and Experimental Science*, vol. III (New York: Columbia University Press, 1934).
Leibniz appreciated Swinehead’s efforts at quantifying problems in physics – though he thought them unsuccessful -- rather than indulging in the traditional methods of scholastic disputation. See CO 177; 340.

33. Leibniz repeats his claim that the basis of binary arithmetic (0 and 1 alone and in various combinations) is further evidence for Creation. (See Letter H, n. 19.) Besides validating his own discovery, the binary basis of Fu-Hsi’s hexagrams demonstrates that the doctrine of Creation was understood by ancient Chinese thought, before or apart from any biblical revelation. Such wisdom testified, for Leibniz, to the logical acuity and intelligence of the ancient Chinese, qualities that would raise European esteem for Chinese thought and civilization. It further would enable the missionaries to claim that by converting to Christianity, the Chinese were simply

returning to their own ancient sources and traditions. Thus the Jesuits, for example, could accommodate these traditions in their missionary efforts. On the accommodationist position of the Jesuits, see D. Mungello, *Curious Land: . . ., Chapters VIII, IX and X*,

34. The number 7 is considered magical or sacred in the biblical tradition because of the account of Creation in the Book of Genesis (II: 2-3): “And on the seventh day God finished His work which He had made; . . and God blessed the seventh day and hallowed it.” Binary arithmetic ratifies this notion, since the number seven (or septenary) is written with three 1’s – which is an intimation of the Trinity. While Leibniz noted this “confirmation” of the Doctrine of the Trinity in the number symbolism of binary arithmetic, it is doubtful, as a rationalist and mathematician, that he gave much credence to all this numerology. He most likely volunteered such “insights” because he thought that such an approach might increase the esteem for Christianity in the eyes of the Chinese.

35. Müller edited a Latin edition of Marco Polo’s travel accounts to the Orient in 1671. For more details and bibliography on Müller, see Letter B, n. 8.

36. On Grimaldi, see Letter B, n. 7.

37. Müller had been in charge of the Oriental collection in Berlin of the Elector of Brandenburg, Friedrich Wilhelm (1640-1688) -- known as the “Great Elector.” On the role of Electors in the Holy Roman Empire, see n. 1 of Letter H, n. 1.

38. On Golius, see Letter H, n. 52.

39. Leibniz gives the full title of this work in the postscript to this letter that he probably added when he returned to Hanover from Berlin in early June.

40. (The note continues in context of the catalogue mentioned.)

41. By denying any connection between Egyptian hieroglyphs and Chinese characters, Leibniz effectively undercuts a key argument of proponents of Ancient Theology. See D. Cook & H. Rosemont, "Leibniz, Bouvet, The Doctrine of Ancient Theology and The Yi-Jing, Das Neueste Über China, p. 135. The most celebrated proponent of the Egypt-China connection was Athanasius Kircher; on Kircher, see Letter E, nn. 11-12.

42. Horapollo= Horus Apollo=Orus or Orion. Orus was a Greek grammarian and rhetorician from Thebes, Egypt who lived in the 5th Century C.E. He was the teacher of Proclus, the famed neo-Platonist, and author of a partially extant work on Greek etymology.

43. Ammianus Marcelinus, a Roman historian of Greek origin, lived in the 4th century C.E.. As a soldier, he traveled throughout the Roman Empire, Asia Minor and Egypt. He wrote a work called *Rerum gestarum libri XXXI*. There is a 1474 Rome edition of books XIV-XXVI by Paulus Sabinus (The first 13 books are not extant.). The later 1533 editions (in Augsburg and Basel) added books XXVII-XXXI. In chapter XVII of this book, Marcelinus offers an explanation of the hieroglyphic characters on an Egyptian obelisk set up at Rome in the Circus Maximus (§6). Later in this chapter (§§18-23), a detailed Greek translation is given of some of the hieroglyphs on this obelisk.

44. Pierus Valerianus=Giampetro Valeriano (1477-1558), Italian scholar whose patrons included Popes Julian II, Leo X and the Medicis. He wrote *Hieroglyphica, sive de sacris Ægyptiorum...* (Basel 1566 [first part only]; Frankfurt, 1678).

45. On Kircher, see Letter E, nn. 11-12.

46. Laurentius Pignorius=Lorenzo Pignoria (1571-1631), Italian scholar and antiquarian, wrote a major work on Egyptian hieroglyphics, that was republished and renamed several times. Leibniz refers to it by its third title: *Mensa Isiaca, qua sacrorum apud Ægyptios...* (Amsterdam: A. Frisii, 1669).

47. On Thevénnot, see n. 17 above. In his *Relations de divers...* mentioned above, there is a selection entitled *Relation du Mexique et de la Nouvelle-Espagne, par Thomas Gage*, that was published separately (Paris: T. Moette, 1696). It is a folio volume with a title page and 58 pages. The first 46 pages are glyphs that Gage then carefully explains in the last dozen pages. These latter pages are separately entitled, *Histoire du Mexique par Figures expliquées en langue mexicaine, et depuis en langue Espagnole*.

48. Father Carlo Mauritio Vota, S.J. (1629-1715) was papal legate to Warsaw and Confessor to the Polish King. Leibniz corresponded with Vota during this period (1698-1709), particularly on matters pertaining to Russia and China. Vota had been head of the Jesuit house in Moscow from 1684 to 1689.

49. This is probably an account of his unsuccessful attempt to return to China in 1691 by an alternate overland route, since “having Muscovy closed to him,” Grimaldi tried “to go from Persia through the lands of the Usbek Tartars and Bokhara onward to China,” but found it too dangerous. Preface to the *Novissima Sinica*, A IV vi, p. 406; CR 57. See also n. 76 below.

50. The *Germaniae antiquae libri tres*, by Philipp Cluverius=Philipp Clüver (1580-1623), German geographer and historian, was republished several times in the early 17th-century in Leiden, where he was active.

51. Leibniz is not willing to accept Bouvet's theory that “the ancient Chinese had the true religion,” but nevertheless recognizes the value of his approach on weaker, pragmatic grounds. See D. Cook & H. Rosemont, Jr., Leibniz, Bouvet, The Doctrine of Ancient Theology and The Yi-Jing, Das Neueste Über China, pp. 135-136.

52. One can see why Leibniz might strike this passage – not wanting to offend his correspondent's Catholic sensibilities [re: damnation of pagan babies and the belief that there is no salvation outside the Church] but it is revealing.

53. Leibniz is referring here to Rosamund von der Asseburg (1672-1727), who appeared on the scene (along with her sisters) in the fall of 1691, making a strong impression on the court in Hanover. Leibniz is quite skeptical of her claims and explains her behavior in naturalistic, psychological terms. He nevertheless says she should be left alone (he likens her to a rare "cabinet piece") on pragmatic grounds, since her activities and visions have positive results. (A I vii: 34-35). For more on Leibniz's attitude toward such enthusiasts (Schwärmer), see D. J. Cook, "Leibniz on Enthusiasm," in Leibniz, Religion and Mysticism, eds. A. Coudert, R. Popkin & G. Weiner (Dordrecht: Kluwer, 1998), pp. 107-135. On Asseburg in particular, see *ibid.*, pp. 121-123. As long as such mystics did not undermine or disturb the public order, Leibniz's attitude was quite open. For an English translation by her foremost follower, see J. W. Petersen, *A Letter to Some Divines... with an exact account of what God hath bestowed upon a noble maid* (London: J. Whitlock, 1695).

54. Caelius Secundus Curio (1503-1569), Italian writer and philologist, wrote *De Amplitudine beati regni Dei Dialogi...* (Basel 1554; Frankfurt 1617) that argued for the fullness of God's salvation. He believed that pagans who observed the tenets of natural law would be saved. Early in his life, he defended Lutheran and Reformed doctrines and eventually found refuge in Switzerland from the Inquisition as a Professor of Literature at the University of Basel.

55. Apparently Leibniz forgot the date of the eclipse he was referring to and planned to insert it later. Plotting the meridian (a line encircling the globe that passes through the poles and any given point between them) was a matter of ongoing scientific concern in the 17th & 18th centuries. On efforts to plot the prime meridian through Paris, see Leibniz's first letter to Bouvet, n. 29. Plotting longitudinal lines (expressed in degrees of a 360° circumference of the earth or minutes of a 24-hour clock) off a prime meridian (e.g., Greenwich) was of paramount importance for proper navigation and map-making. On the fascinating search for proper longitudes, see the book by Dava Sobel, Longitude.

56. Leibniz is referring to the legendary ruler Zhuanxu (2513-2435 B.C.E.). See WB 738, n. 52.

57. Until this letter, Leibniz had never asked Bouvet directly about the Jews in China, but had asked Verjus and Gobien earlier about their Bible. Leibniz, as well as his Jesuit correspondents, were intrigued by the existence of Jews in China for several reasons and later tells Bouvet that "it is important to thoroughly examine their doctrine and their rites." (Letter N) First and foremost, Leibniz (and Christian Europe) was eager to verify the text of the Old Testament, since its prophecies are the prime source for the Mosaic chronology of the world. Leibniz, like many others, was intent on harmonizing such accounts with contemporary astronomical and geological findings. For these reasons, it was "important to make at least a copy of their beginning of Genesis, to see if their genealogy of the Patriarchs perhaps agrees with that of the Septuagint text." (Letter to Verjus, 1 January 1700 [A I xviii, p. 273]) Second, as a defender of orthodox Christianity, Leibniz hoped that earlier Hebrew texts of the OT from the pre-Christian era might contain "books or passages that the Jews of Europe may have changed or suppressed out of antipathy to Christianity." (Ibid.) Finally, if Jews had been in China from "time immemorial," it would lend credence to the "figurist" theories of Bouvet and others: namely that these Jews were descendants of Noah, the Patriarchs or "The lost Ten Tribes," and thus the source of the divine truths that were embedded in the ancient classic texts of China.
In Letter O (unanswered), his final letter to Bouvet (see n. 2), Leibniz forwards a list of questions about the Jews of China from Daniel Jablonski (1660-1741), the Protestant theologian and Court Preacher in Berlin, who edited an edition of the Old Testament (1699). At about the same time, Leibniz excerpted some materials on the Jews in China (see WB, pp.590-593). There is an extensive literature on the Jews in China, especially on the ancient community in Kaifeng. The most comprehensive work on this topic is by D. D. Leslie, *The Survival of the Chinese Jews: The Jewish Community of Kaifeng* (Leiden: Brill, 1972). The latest effort is *From Kaifeng...to Shanghai: Jews in China*, ed. R. Malek (Nettetal: Steyler Verlag, 2000. See in particular, R. Widamier, "Zur Frage der Juden in China in der Korrespondenz von G.W.Leibniz," *ibid.*, pp. 559-588.

58. The Samaritans are traditionally thought to be the offspring of the local population and the Jews who remain in northern Israel (i.e., Samaria) after the Babylonian exile. The Samaritan Pentateuch, which contains minor but significant differences from the Judaeo-Christian Pentateuch or Five Books of Moses, is the exclusive source for their basic doctrines, observances and traditions. The Samaritan *Book of Joshua* (extant as a much later compilation of materials and written only in Arabic) should not be confused – as Leibniz apparently did -- with the Hebrew *Book of Joshua*, the sixth book of the Hexateuch. The Hebrew text of the Samaritan Pentateuch reached Western Europe only at the beginning of the 17th century. It occasioned great excitement on the part of biblical scholars, who thought (incorrectly it turns out) that it might be an older edition of the Hebrew Bible than the traditional Masoretic text used by (Rabbinic) Jews. *Encyclopaedia Judaica*, XIV, cols. 265-266.

59. The Hebrew script used to transcribe and transmit the Pentateuch (i.e., “square” or “Assyrian” script) was indeed adopted by the Jews in Babylonia and replaced the paleo-Hebrew Script. “The Samaritan alphabet is the only descendant of the early Hebrew Script which is still in use.” *Ibid.*, col. 267. The Samaritan system of vowels and reading accents of the Hebrew text of the Pentateuch is completely different from the Masoretic one. *Ibid.*, XIII, col. 1468.

60. See WB 739, nn. 55-56.

61. On Martini, see Letter H, n. 47. Leibniz is referring here to Martini’s major cartographic work, *Novus Atlas Sinensis* (Amsterdam: J. Blaeu, 1655).

62. On Couplet and his *Confucius Sinarum philosophus* . . . , see Letter E, n. 22. Couplet gives the so-called King Wen (also called the *hou tian* 後天 [post-Heaven] ordering of the *Yi jing* Hexagrams, whereas Bouvet provides the so-called “Fuxi” ordering (also called the *xiantian* 先天 or pre-Heaven ordering).

63. On Golius, see Letter H, n. 52. Standaert mentions three items from “the sales catalogue of his library, which was publicly sold in 1696”: . . . “a ‘thesaurus’ with the explanation of 10,000 characters in Spanish; . . . a Spanish-Chinese vocabulary annotated by Golius; and a Spanish booklet containing the pronunciation of the Chinese characters.” N. Standaert, *Handbook of Christianity in China*, p. 871.

64. Hiob Ludolf, the famed German linguistic scholar, had written to Leibniz in 1698 about the *Hai pian*. But Leibniz had heard mention of this work previously through the questions that he forwarded to Bouvet from Father Kochanski (A I xiv: N. 450, question no. 22). On versions of the *Hai pian*, the so-called “Ocean dictionary”, in Europe, see D. Mungello, *Curious Land. . .*, pp. 217-218, and WB, p. 649-50, n. 16). There has been some confusion about the precise dictionary meant by the name *Hai pian*, since in China there was no single dictionary of that name; instead, a good number of dictionaries from the Ming dynasty had the words *hai pian* in the title, so many that these were later seen to be a class in itself. These were in great measure directly derivative of the early Ming Jingshi haipian zhiyin 經史海篇直音 (Correct pronunciation of an oceanic compendium of [words from] the classics and histories), itself owing directly to a few earlier works from the Jin dynasty and the early Ming whose title contained the words pian hai “篇海”. The Jingshi haipian zhiyin, most often referred to by the abbreviated “Hai pian” was diffused especially through it’s reissue with minor emendations by Zhang Fu 章黼 (fl. ca. 1544?), titled Xinjiao .... 新校.... (Newly collated....), which went through various printings beginning at least from 1544; several of these still exist. The “newly collated” version was immense, containing 55665 separate character entries, many of them simple graphic variants, but the greater importance of this work is that it was one of the very first lexicons since the Han to organize characters by groupings based on a set of recurring constituent graphic

elements instead of by pronunciation, serving as a direct influence on the development of the dictionary format that was “standardized” by the system of 214 headings (“radicals” or semantic determinatives), formalized by Mei Yingzuo 梅膺祚 (1570-1615) in his grand lexicon Zi hui 字匯 (ca. 1615; Compendium of characters) that contained 33179 characters. This was the model for the great Kangxi Dictionary (Kangxi zidian 康熙字典) of 1716 that contains 47035 characters under the 214 graphic categorical headings. See Endō Yuriko 遠藤由里子, “Kaihen shohon kō” 海篇諸本考, Baikō jogakuin daigaku ronshū 梅光女学院大学論集 22 (1989): 31–57; Ōiwamoto Kōji 大岩本幸次, “Mindai Kaihenrui jishogun ni kansuru nisan no mondai” 明代「海篇類」 字書群に関する二三の問題, Tōhoku daigaku Chūgokugogaku bungaku ronsō 東北大学中国語学文学論集 4 (1999): 19-33; and Yang Zhengye 楊正業 “Haipian chengshu niandai kao” 海篇成书年代考, Cishu yanjiu 辞书研究 (2005.1): 174-176.

65. See Letter H, n 52.

66. Leibniz is already suspicious that he is not receiving everything that Bouvet is sending out to him through Paris. His suspicions soon become realized, since Leibniz never again receives an answer from Bouvet, including his very long letter addressed to Abbe Bignon (15 September 1704) which was to be forwarded also to Leibniz (see C. von Collani, *Eine Wissenschaftliche Akademie...*, pp. 26-27).

67. Leibniz adds a note to Fontaney at the end of this letter. As is apparent from his remark immediately below, Leibniz thought that Fontenay was still in China, not knowing he had left China on the Amphitrite in December 1702. He arrived in Europe at the end of 1703 on an English ship and does finally respond to Leibniz in a letter from Paris, 13 June 1704 (WB: 436-439).

68. See his letter to Leibniz of 15 September 1701. (WB 446-451).

69. Strabo (64 B.C.E.?—24 C.E.?) was a noted Greek geographer and historian. No such passage can be found in his works. The German editors of the correspondence think it is likely that Leibniz was referring to Pliny the Younger's *Naturalis historia*, 8. 173 (WB 740, n. 64).

70. The famous stele discovered near Xi'an (Sian) in 1623/25 (it is in the Beilin Museum at Xi'an) demonstrated that the Nestorian or Assyrian Christians were active in China during the Tang Dynasty (618-907 C.E.). The inscription on the stele is dated at 781 C.E. and "consists of 1800 characters and is divided into several parts." N. Standaert, *Handbook of Christianity...*, p. 3. The Chinese text from the stele presents the theology, history and prayers of the Nestorians and was reproduced in Kircher's *China Illustrata* of 1667. For the latest information on the stele and the controversies about its authenticity, where and when it was discovered, etc., see

*ibid.*, pp. 3-4; 12-15 *et passim*. For a detailed discussion on the "Nestorian monument" as well as an excellent reproduction of its text as it appeared in Kircher, see D. Mungello, *Curious Land...*, pp. 164-172.

71. Sent by Bouvet to Leibniz along with Letter E.

72. Such a volume cannot be found any longer in the Staatsbibliothek of Berlin (WB 740. n. 67).

73. Bouvet refers to this project of a Manchu-Chinese dictionary in Letter F, p. 5.

74. As of the writing of this letter, no Papal bull had yet been issued on whether or not the traditional Confucian religion should be considered a form of atheism or idolatry that was incompatible with Christian doctrine. Leibniz repeatedly took the Jesuit side, which thought that Christianity could accommodate – theoretically and practically – such Chinese beliefs as well as their on-going sacrificial rites. See Letter B, n. 1, on the so-called “Rites Controversy.” The Church finally came down on the side of the opponents of Leibniz and the Jesuits in the next decades with the Papal bulls *Ex illa die* (1715) and *Ex quo singulari* (1742), which ruled against their interpretation of the Chinese rites. See D. Mungello, *Curious Land . . .*, p. 357. Leibniz’s strongest and longest effort to defend the accommodationist view is his *Discourse on the Natural Theology of the Chinese*, written in the last year of his life (1716). In the *Discourse* Leibniz uses the very same Chinese materials from the anti-accommodationist treatises of Longobardi and St. Marie—received from Nicholas Remond in the autumn of 1715—to justify his opposing view. For details on the relevant correspondence and a critical edition of the *Discourse*, see LP. For an English translation and study of the *Discourse*, see CR.

75. By issuing the Edict of Toleration in 1692, allowing Christianity to be practiced in China, the Kangxi emperor in effect sided with the Jesuits, though he had only affirmed in a neutral vein that Christianity could be tolerated since it was not incompatible with the religion of the Chinese. In 1700 the emperor responded more positively by approving certain specific Jesuit positions. “The Jesuits in Beijing had solicited the Kangxi emperor’s opinion on the three major issues of their ‘accommodated’ Christianity (tian, “Heaven,” denotes the supreme Being; the ancestor cult is an expression of gratitude; the cult of Confucius is a civil non-superstitious ceremonial), and the Emperor had affirmed that this indeed was the right interpretation.” N. Standaert, *Handbook of Christianity . . .*, p. 498. Le Gobien had communicated the details of this successful petition to the Emperor in a letter to Leibniz from Paris, 10 November 1701 (WB 378-381). A text of these imperial opinions plus other documents, entitled (perspicuously enough) *Brevis Relatio eorum quae spectant ad declarationem Sinarum Imperatoris Kam Hi circa coeli, Cumfucii et avorum cultum, datum anno 1700* (Augsburg and Dillingen: Bencard,1703), later were sent to Rome. This effort was rebuffed as Rome was offended that a pagan dared pass judgments on

matters of Christian faith and doctrine. (See WB 729 and C. von Collani, *Eine wissenschaftliche Akademie für China*, p. 13).

76. Eluth, better known as Ölet, refers to the area known now as the Altai region of Outer Mongolia. It was inhabited at the time by the Ölots and included areas of Chinese Turkistan (Jungaria) and northeastern Tibet. Starting in the late 17th Century, the Ölots or Western Mongols were gradually brought under the control of the Manchus, who by the end of the 18th century dominated all of Outer Mongolia. See O. Lattimore, *Inner Asian Frontiers of China* (Boston: Beacon Press, 1967) pp. 78-79; 86-87. Since the overland route to China through Russia was closed to the Jesuits (see Letter H, n. 15), Europe hoped that an alternate route could be found by Persia, or northern India (the Moghul Empire), through Turkistan ("the Usbeks") to China. However, as Leibniz here suspects, such a route never materialized.